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Textual Analysis in Accounting: What's Next?*

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ABSTRACT

Natural language is a key form of business communication. Textual analysis is the application of natural language processing (NLP) to textual data for automated information extraction or measurement. We survey publications in top accounting journals and describe the trend and current state of textual analysis in accounting. We organize available NLP methods in a unified framework. Accounting researchers have often used textual analysis to measure disclosure sentiment, readability, and disclosure quantity; to compare disclosures to determine similarities or differences; to identify forward-looking information; and to detect themes. For each of these tasks, we explain the conventional approach and newer approaches, which are based on machine learning, especially deep learning. We discuss how to establish the construct validity of text-based measures and the typical decisions researchers face in implementing NLP models. Finally, we discuss opportunities for future research. We conclude that (i) textual analysis has grown as an important research method and (ii) accounting researchers should increase their knowledge and use of machine learning, especially deep learning, for textual analysis.

Keywords: textual analysis, machine learning, deep learning, textual data, content analysis, text

L'analyse textuelle en comptabilité : quelle est la prochaine étape?

RÉSUMÉ

Le langage naturel est une forme essentielle de communication d'entreprise. L'analyse textuelle correspond à l'application du traitement du langage naturel (TLN) aux données pour extraire ou mesurer automatiquement des informations. Les auteurs examinent les publications des principales revues de comptabilité et décrivent la tendance et l'état actuel de l'analyse

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textuelle en comptabilité. Ils classent les méthodes TLN pratiquées dans un cadre unifié. Les chercheurs en comptabilité ont souvent utilisé l'analyse textuelle pour mesurer le sens, la lisibilité et la quantité des informations fournies, pour comparer les informations afin d'établir les similarités ou les différences, pour identifier les informations prospectives et pour détecter les thèmes. Pour chacune de ces tâches, les auteurs expliquent l'approche conventionnelle et les approches plus récentes, qui sont basées sur l'apprentissage automatique, en particulier l'apprentissage profond. Ils discutent de la manière d'établir la validité d'exécution des mesures basées sur le texte et des décisions typiques auxquelles les chercheurs sont confrontés lors de la mise en œuvre des modèles de TLN. Enfin, les auteurs discutent des possibilités de recherches futures. Ils concluent que (i) l'analyse textuelle s'est développée en tant que méthode de recherche importante et (ii) les chercheurs en comptabilité devraient accroitre leurs connaissances et leur utilisation de l'apprentissage automatique, en particulier l'apprentissage profond, à des fins d'analyse textuelle.

Mots-clés : analyse textuelle, apprentissage automatique, apprentissage profond, données textuelles, analyse de contenu, texte

1. Introduction

In his presidential address to the American Economic Association on January 7, 2017, Robert J. Shiller emphasized the importance of narratives in understanding the motivations and economic impact of human behavior (Shiller 2017). He defined narratives as "human constructs that are mixtures of fact and emotion and human interest and other extraneous detail that form an impression on the human mind" (Shiller 2017, 973). Text constitutes much of a narrative. Analyzing a large quantity of textual data was labor-intensive until *natural language processing* (NLP) methods became available.[1,2] Textual analysis is the application of an NLP method to textual data for automated information extraction or measurement. Over the past two decades, textual analysis has been increasingly conducted in research in social science.[3] In this article, we describe the trend and current state of textual analysis in accounting research, organize available methods in a unified framework, explain their current and potential applications in accounting, emphasize the importance of construct validity, provide implementation guidance, and discuss future research opportunities.

We review studies published in top accounting journals from 2010 through 2021 and identify 207 studies that use textual analysis (see details in Appendix 2). Over this period, the number of textual analysis publications has grown substantially, from an annual average of 6 in the period of 2010–2013 to 14 in 2014–2017 to 32 in 2018–2021. In the most recent period of 2018–2021, 15.6%, 12.7%, and 10.3% of the total publications in *Journal of Accounting Research*, *Review of Accounting Studies*, and *The Accounting Review*, respectively, use textual analysis, suggesting that textual analysis has quickly become popular for accounting research.

Textual analysis plays various roles in accounting research: 52.7% of the textual analysis publications construct a variable directly from a document and use the variable as the dependent or explanatory variable for hypothesis testing, 29.9% extract targeted textual information (e.g., through keyword search or regular expressions) and construct a variable from the extracted data for subsequent analysis, 5.8% construct a text-based control variable, 2.4% produce text-based variables as predictors of an outcome of interest, and 9.2% use text-based measures for

1. Before the advent of textual analysis, accounting researchers used human coding to analyze textual data. The accounting literature refers to such analysis as "content analysis." Content analysis is still used for extracting complex information from a relatively small set of documents. See Jones and Shoemaker (1994) and Cole and Jones (2005) for the literature reviews.
2. We italicize a term the first time it appears in our article and provide a functional definition in Appendix 1.
3. See Schwartz and Ungar (2015), Gentzkow et al. (2019), Berger et al. (2020), Loughran and McDonald (2020), and Colón-Ruiz and Segura-Bedmar (2020).

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other purposes. These observations suggest that textual analysis plays a primary role in research designs and that accounting researchers typically use textual analysis to construct a measure.

Given the growth and importance of textual analysis in accounting research, we provide an overview of available NLP methods in a unified framework and explain how NLP has advanced in computer science. The framework is based on the responses to the following three questions for any NLP process. How are textual data represented as inputs to the NLP model? What model is used to transform these inputs to outputs? What output is generated from the NLP process? We discuss two approaches to text representation: (i) the *bag-of-words* (BOW) approach, which represents textual documents as vectors of word counts, and (ii) the *word embedding* approach, which captures semantic relationships among words and reduces textual dimensionality. We group NLP models into four categories: (i) simple transformations, (ii) text comparisons, (iii) traditional *machine learning*, and (iv) *deep learning*. The type of output generated from an NLP process depends on the research objective.

We then cover in detail the NLP applications in accounting. Accounting researchers have commonly used textual analysis to measure disclosure sentiment, readability, document similarity, forward-looking information, and topics. For each of these tasks, we explain the conventional NLP methods and then discuss newer methods that may improve measurement of the construct of interest. For example, the conventional NLP method for measuring disclosure sentiment counts positive and negative words in a document using a predetermined sentiment dictionary (the combination of BOW and use of a dictionary is typically referred to as "the dictionary approach" in the accounting literature). This method is simple but, as we discuss in section 2, has limitations. Traditional machine learning methods provide alternatives, which are often used along with the BOW text representation. Deep learning provides newer methods, which are often used with the word embedding text representation.

Given that textual analysis is typically used to construct new measures for hypotheses testing, we next highlight the importance of establishing the construct validity of text-based measures. We introduce the multiple facets of construct validity and provide practical examples of how researchers can establish the construct validity of their measures.

The next part of our article explains common decisions researchers face when implementing NLP methods, for example, data acquisition and preprocessing, model selection, and implementation decisions for supervised machine learning. We also outline available resources (e.g., libraries, code, and textual measures) that may help researchers begin NLP implementations.

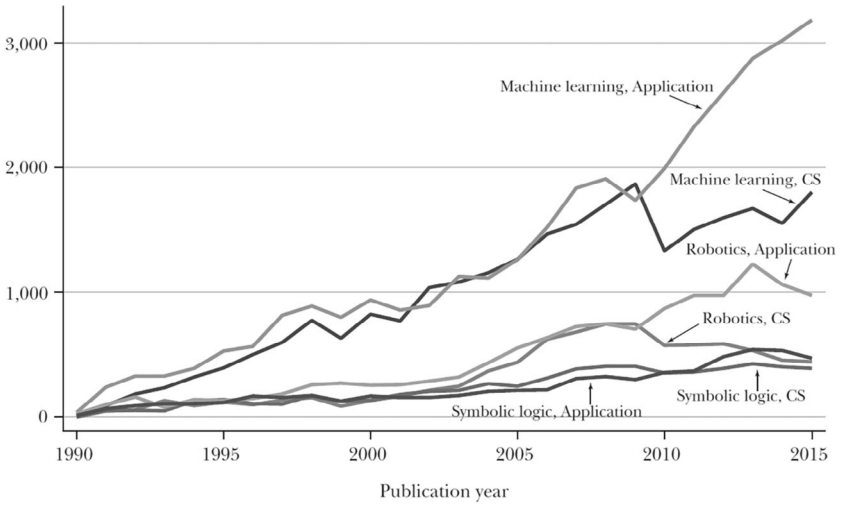
The final part of our article discusses two avenues for future research. First, we discuss supervised deep learning for scaling up the work that can be performed by humans on small samples. Using this tool, researchers can address new research questions that were out of reach because of resource constraints or limitations of traditional NLP methods. The second avenue is topic modeling. Textual analysis studies have primarily focused on syntactic features (e.g., number of syllables and count of independent words) or simple semantic features (e.g., positive vs. negative words) of corporate disclosures. Accounting research has just started to detect richer semantic features, such as disclosure topics. Newer NLP methods can help accomplish these goals.

Our survey makes three contributions to the accounting literature. First, we highlight textual analysis as a research method. Textual analysis can help researchers address their questions by providing a measure that better reflects the underlying construct (e.g., using the content of a disclosure instead of just its existence) and better illuminating proposed causal relationships (e.g., showing the linkage of actual information disclosure in a spillover study). Textual analysis can enable researchers to explore questions that were previously not executable because of measurement difficulties.

Second, our article calls attention to doctoral training. Given the advances in NLP and computing power and the availability and large volumes of textual data, the number of accounting

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Figure 1 Trends of machine learning applications



Notes: This figure is taken from Agrawal et al. (2019, figure 1). It depicts the number of publications in application journals that use technologies from three areas of artificial intelligence—machine learning, robotics, and symbolic logic. The graph also shows the number of publications in computer science (CS) for developing the above technologies. For example, the line at the top represents the number of publications that apply machine learning models. This line shows a smooth, continued growth from 1990 to 2009 and then a substantial growth since 2009.

studies that use textual analysis will continue to grow. Future graduates of accounting doctoral programs will be expected to possess textual analysis skills. To our knowledge, most programs do not have resources for training their students in textual analysis and even fewer have exposed their students to machine learning. Our article may encourage young researchers to invest more time developing textual analysis skills and influence doctoral programs to either locate available resources within their institutions or outsource this training.

Last, we provide a more systematic and complete coverage of available NLP methods and their applications than previous surveys. The first survey, F. Li (2010a), was written when textual analysis in accounting was in its infancy. Although the first accounting study using textual analysis was published in 2004 (Butler et al. 2004), only seven such studies were published or forthcoming prior to F. Li (2010a).[4] In the second survey, Loughran and McDonald (2016) focus on the dictionary approach of textual analysis. Since their survey, there have been significant advances in NLP (see Figure 1; Agrawal et al. 2019; LeCun et al. 2015). These advances not only address some methodological issues raised by Loughran and McDonald (2016) but also open new avenues for future research. It is important for academic researchers to see various NLP methods in an overarching framework and understand their pros and cons. We expect our systematic coverage of NLP methods to help beginners of textual analysis get started, researchers with some experience explore alternative (perhaps superior) methods, and reviewers better evaluate textual analysis.[5]

4. See F. Li (2008, 2010b), Henry (2008), Kothari et al. (2009), Feldman et al. (2010), S. V. Brown and Tucker (2011), and Lehavy et al. (2011). F. Li (2010a) also covers working papers, some of which were subsequently published.
5. Even though a few recent textual analysis surveys exist in other disciplines, such as Schwartz and Ungar (2015), Gentzkow et al. (2019), Berger et al. (2020), and Loughran and McDonald (2020), none provides an overarching framework and none discusses deep learning to the extent we do.

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2. An overview of NLP methods

An NLP process has three components: (i) representation of textual data as inputs, (ii) a model that transforms inputs to outputs, and (iii) outputs. An example of inputs is the number of occurrences of each word in a document. An output is a measure to capture the underlying construct of interest (e.g., disclosure sentiment).

We organize available NLP methods by the two primary components of an NLP process—the text representation approach and the model type. Table 1 presents the methods along these two dimensions, with examples of the construct that the output intends to capture as well as the first accounting application (of which we are aware) in parentheses. Alternative NLP methods may be available for capturing the same construct. For example, both dictionary-based and machine learning methods can be used to measure disclosure sentiment.

The current section discusses the text representation approaches and model types. Section 3 discusses the NLP applications that capture the constructs of common interest to accounting researchers. Because newer text representation methods use the architecture underlying some NLP models, we first discuss the NLP models and then the text representation approaches.

2.1 NLP models

NLP models fall into two broad categories: (i) rule-based and (ii) data-based transformations. The former transforms inputs to outputs using manually developed rules (e.g., a dictionary for climate risk). Once the rules have been developed, the transformation is straightforward. However, it can be time-consuming to develop these rules, and the rules might not scale well to complex situations such as distinguishing risk disclosure from forward-looking information. We further classify rule-based transformation models into simple transformations versus text comparisons.

Data-based transformation models let the machine learn patterns in *training* data without human intervention during the fitting of the model and are referred to as "machine learning." Researchers have extensive control over the selected *features* (individual measurable properties or characteristics) used by the model and must carefully tune the *hyperparameters*—that is, parameters that need to be set by researchers to control the learning process of the model. Many machine learning models have been adapted to NLP and some are designed for NLP, such as topic modeling. Machine learning can be classified into (i) *supervised learning*, where the output variable in the training data has been annotated (labeled) for the machine to learn the relationships between inputs and outputs, and (ii) *unsupervised learning*, where there are no labeled outputs, and it is the machine's job to discover latent outputs (e.g., semantic clustering).[6] We separate traditional machine learning models, which have been developed in computer science and applied in other disciplines for decades, from deep learning models, which have been developed and applied only recently.

2.1.1 Simple transformations

These models take inputs and return them as outputs with minimal transformation. Popular examples include (i) keyword search to identify or extract the information relevant to a research question; (ii) count of words, sentences, or articles that contain certain keywords; (iii) identification of complex words, long sentences, or other writing elements (e.g., readability); and (iv) classification of certain types of words (e.g., counts of negative words). These transformations are relatively easy to execute.

6. Semi-supervised learning falls between the two categories. For example, for sentiment classification, the machine is given the sentiment for some sentences (supervised learning) but uses word embeddings to represent words (unsupervised learning).

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TABLE 1
Textual analysis methods, tasks, and application examples

Model			Text representation	
Type	Subgroup	Model	Bag of words	Word embedding
Simple transformation			Keyword search (Butler et al. 2004) Disclosure quantity (Frankel et al. 2010) Readability (F. Li 2008) Sentiment (Kothari et al. 2009) Forward-looking information (F. Li 2010b)	
Text comparisons		Cosine similarity model	Disclosure similarity (S. V. Brown and Tucker 2011)	Disclosure similarity (Gaulin and Peng 2021)
Traditional machine learning approaches:				
Naïve Bayes			Sentiment (F. Li 2010b)	
KNN				Industry classification (Song 2021)
Text regression	Linear	Penalized regression (e.g., Lasso)	Investment prediction (Basu et al. 2022)	
	Nonlinear	CART (e.g., random forest)	Sentiment (Frankel et al. 2021)	
		Support vector machine	Fraud prediction (Purda and Skillicorn 2015)	
Topic modeling		Unsupervised (e.g., LDA)	Disclosure topics (A. Huang et al. 2018)	
		Supervised (e.g., sLDA)	Credit risk (Donovan et al. 2021)	
		Embedded topic model (e.g., ETM)		
Deep learning approaches:				
		ANN		
		CNN		Forward-looking information (S. V. Brown et al. 2021)
		RNN (e.g., LSTM)		Disclosure objectivity (De la Parra 2020)
		Transformers (e.g., BERT)		Sentiment (A. Huang et al. 2022)

Notes: Abbreviations used in the table are as follows: artificial neural network (ANN), bidirectional encoder representations from transformers (BERT), classification and regression trees (CART); convolutional neural network (CNN); embedded topic model (ETM), *K*-nearest neighbors (KNN), latent Dirichlet allocation (LDA), long short-term memory (LSTM), recurrent neural network (RNN), and supervised LDA (sLDA).

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The interpretability of the outputs, however, can be challenging because interpretation relies on intuition, psychology, or other sources. For example, *document* length could proxy for disclosure quantity or for obfuscation if the underlying information content is held constant.[7] More syllables per word could reduce readability, yet rules are needed to handle words that have greater-than-average syllables but are highly readable in certain instances, such as "company" or "industry" to financial statement users.

2.1.2 Text comparisons

Some models go beyond a simple transformation of the inputs without using any statistical or machine learning models. Text comparison models fall into this category. For example, the *cosine similarity* model using the *vector space model* (VSM) represents each document with an *n*-dimensional vector and uses the angle between two vectors to measure the similarity of two documents.[8] This measure was originally designed for information retrieval (i.e., finding documents similar to a query) but can be used for other purposes, such as plagiarism detection and quantifying how much a disclosure has changed over time. If the vectors represent features other than simple word counts, such as topic mix, cosine similarity can still be calculated and interpreted.

2.1.3 Traditional machine learning

We discuss four subgroups of traditional machine learning models: (i) *naïve Bayes* (NB), (ii) *K-nearest neighbors* (KNN), (iii) text regressions and decision trees, and (iv) topic modeling. The first three categories are supervised algorithms, and the last category is typically unsupervised.

NB is a probabilistic supervised learning classifier based on applying Bayes' theorem.[9] For textual data, the unit of classification can be sentences, paragraphs, or the entire document. Each unit is reduced to a list of words with the frequency of each word. The training data include labeled classifications; the algorithm estimates the weights of the words that best yield the true classifications. These weights are then used to classify unseen data. NB has been a popular machine learning method for textual analysis, but its performance may be limited by the assumption that the features used to predict the classification categories are independent.

KNN is a supervised learning algorithm that can be used to produce categorical or continuous outputs (Aggarwal 2018, 133–142). "Nearest neighbors" are the data points in the training set that have minimum distance in the feature space from a new data point. "K" is the number of such data points that the researcher considers in implementing the algorithm. The model is a "lazy" learner because it does not learn a discriminative function from the training data. When predicting the output value for a new data point, the model considers all the data points in the training set and finds the new data point's KNN from the feature space and their output values. For classification tasks, the output value for the new data point is the majority of KNN labels. For predicting a continuous value, KNN takes the mean or median of KNN output values. KNN is intuitive but computationally expensive because it scouts the entire training set for each new data point.

Text regressions are regressions with a mechanism to address the high dimensionality of textual data. The use of thousands of unique words in textual data usually results in many features as inputs to the NLP model. The high dimensionality not only makes common regressions, such as OLS, infeasible when the number of parameters to be estimated exceeds the number of

7. In NLP, a "document" is a general term referring to a text of any length, ranging from phrases to books.
8. Jaccard distance or Levenshtein distance are two popular approaches for measuring similarities of shorter text, such as company name matching (Schütze et al. 2008). The former refers to the number of common words in two texts scaled by all words; more words in common indicate greater similarity. The latter refers to the number of single-character edits required to change one word into the other; fewer edits imply greater similarity.
9. The multinomial version of NB is more appropriate for NLP than the traditional Bernoulli version (see Aggarwal 2018, 123–132 for an in-depth discussion).

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observations but also contributes to *overfitting*. The latter problem occurs when many features help fit the existing sample well, but the model performs poorly in predicting out of sample (Varian 2014). One solution is *regularization*—that is, adding a penalty term in the loss function to reduce the magnitudes of certain parameters (called "weights" or "coefficients"). Two popular penalized linear regressions are *ridge*, which selectively shrinks coefficients, and *lasso*, which not only shrinks some coefficients but also reduces some to zero and therefore produces parsimonious models. Lasso was developed later and has overtaken ridge in popularity (Hindman 2015).

There are two types of nonlinear text regressions. The first belongs to the decision or regression tree family, called *classification and regression* trees (CART). These learning tree models partition data into two groups based on the available best-predicting features and further partition each group based on the remaining best-predicting features. The process stops along a given branch when none of the remaining features would further improve the prediction. Consequently, the final prediction is the machine's classification output. Most tree models can be influenced by extreme observations, are path-dependent, or result in overfitting. The most recent and popular model in this family is *random forest* (RF), which overcomes these weaknesses by randomly sampling across both observations and features to create an ensemble of trees (Hindman 2015).

The second type of nonlinear text regressions uses a *support vector machine* (SVM), which presents training data as data points in high-dimensional space and finds a hyperplane to separate the labeled categories with the highest possible margin—that is, the distance between the hyperplane and the observations closest to the hyperplane. New data points are then classified depending on which side of the hyperplane they land. SVM can capture a linear relationship between the features and the output. If a hyperplane is unavailable in the original feature space, researchers can transform the features into another space using a kernel function, and the resulting classifier is nonlinear with respect to the original feature space. SVM is semiparametric in the sense that it estimates parameters but allows for a family of structural possibilities (Cui and Curry 2005). When the output variable is continuous, a variant of SVM called support vector regression (SVR) is used. SVR has a regularization mechanism to handle high-dimensional inputs.

Topic modeling is typically unsupervised. "Topic" is a latent attribute (output) of textual data that relies on the tendency of words to co-occur. Intuitively, topic discovery is searching for clusters of related words and reduces the dimensionality of the input data into those clusters (Schwartz and Ungar 2015).[10] There have been several topic discovery approaches, but the most popular model is latent Dirichlet allocation (LDA), as indicated by over 43,000 Google Scholar citations of Blei et al. (2003). The model provides probabilistic descriptions of the distribution of topics in a document and the distributions of words affiliated with a topic. Because LDA is a probabilistic model for which the number of topics and any priors must be specified upfront, topics generated and interpreted by one researcher might not be exactly the same as those generated and interpreted by another. To facilitate subsequent analysis, researchers usually interpret and label a topic based on the most frequently occurring words for that topic. Topics can also be used without interpretation, such as examining the topical similarity of two documents.

2.1.4 Deep learning

Traditional machine learning remains effective in certain situations but can easily reach a point where additional complexity in the problem becomes very challenging to overcome. This difficulty arises from three sources. First, even though some traditional machine learning models can learn nonlinear relationships, learning can become difficult or even impossible when the relationships are complex. Second, traditional machine learning requires researchers to manually identify

10. Topic discovery can also be viewed as deriving an open vocabulary from data, where the vocabulary consists of topics instead of words or phrases (Schwartz and Ungar 2015). Words and phrases belong to a closed vocabulary (i.e., an existing dictionary).

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features as inputs to the model. For example, if researchers want to develop a measure of readability using traditional machine learning, they need to supply predefined features, such as the number of syllables and sentence length, as inputs to the model. Third, researchers often train traditional machine learning models from scratch and the training may require substantial data.

Deep learning can overcome the above challenges. It mimics the human brain by creating an *artificial neural network* (ANN) in which artificial neurons are linked into a network connected by edges between the neurons. Deep learning goes beyond classical ANNs by using three or more hidden layers (which makes the network deep) and fewer nodes per layer for better and faster performance even in complex situations. At a conceptual level, deep learning is "a particular kind of machine learning that achieves great power and flexibility by representing the world as a nested hierarchy of concepts, with each concept defined in relation to simpler concepts, and more abstract representations computed in terms of less abstract ones" (Goodfellow et al. 2016, 8). The hidden layers generally represent increasing levels of processing complexity from the first to the last hidden layer. For example, the first hidden layer may "understand" words, and the last hidden layer may "understand" sentiment.[11] To better understand how deep learning works, we next discuss the architecture of ANNs and the evolution of deep learning models.

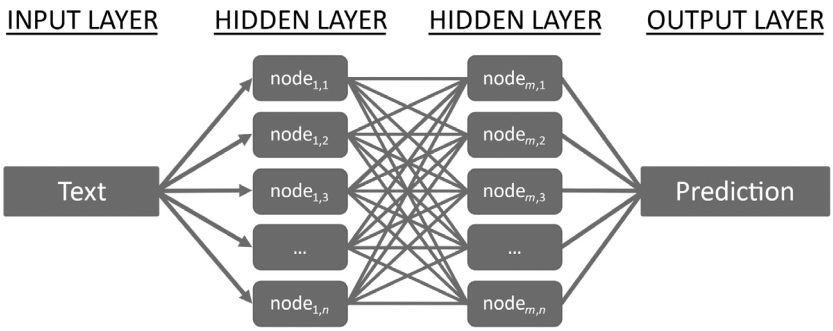
ANNs in their basic form have existed since at least 1980 (and artificial neurons since the mid-1900s) when they were first used for pattern recognition in images (Fukushima 1980). ANNs have at least three layers of neurons: an input layer, one or more hidden layers, and an output layer (see Figure 2).[12] The input layer represents the input data; the output layer produces a set of continuous numbers (e.g., the probabilities of a sentence having positive, negative, or neutral sentiment). These continuous outputs may be left as is or mapped to a category (e.g., "negative sentiment"). The hidden layers learn to perform a mapping of inputs to outputs of the model by adjusting the weights and biases (similar to slopes and intercepts in a traditional regression) at each hidden layer, with the adjustments being based on the chosen activation function and loss function. Researchers choose how many hidden layers are in the model and how many nodes are in a layer—a decision that varies widely depending on the setting, task complexity, and observed performance of the fitted model. Researchers also make other decisions about each layer, such as whether all or a subset of nodes in one layer are connected to all or some of the nodes in a subsequent layer. Initial acceptance of ANNs was slow due to technological limitations and a lack of reliable training techniques.[13]

Early applications of ANNs were concentrated in image processing, where an image is a two-dimensional matrix of pixels. Because these pixels are meaningful only in relation to the surrounding pixels, a technique called a *convolutional neural network* (CNN) was devised to slide a set of small matrices over the image, allowing the immediately surrounding pixels in every direction to form the spatial context for analysis.[14] By using a larger data set and a gradient-based (i.e., derivatives for vectors) optimization method known as *backpropagation*, originally introduced by Rumelhart et al. (1986), LeCun et al. (1998) vastly improved the performance of CNNs such that they could be used commercially for processing bank checks and reading postal addresses. By the mid-2000s, improved training techniques were combined with multiple hidden layers, leading to the first use of the term "deep learning" (Goodfellow et al. 2016, 18).

11. See Meursault (2019) for an example of the use of multiple layers.
12. A logistic regression in machine learning can be considered the simplest neural network with only input and output layers and a sigmoid *activation function* (Sarle 1994). A neural network with a linear activation function and without any hidden layer is a linear model. It is the hidden layers that detect nonlinearities and provide flexibility in learning. From a broader perspective, rule-based transformation models can be viewed as an ANN with only input and output layers and researcher-provided rules (weights) for the activation function, whereas supervised traditional machine learning models replace these rules with weights estimated from training data.
13. Classical ANNs often suffer from the same problem as traditional machine learning—overfitting and difficulty of tuning.
14. These small matrices are called "convolutions" or "filters"; they are typically 3×3 or 5×5 pixels in size.

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Figure 2 Illustration of an ANN



Notes: The model generates a prediction in the output layer from a text document given in the input layer. For example, "Text" could be an earnings announcement press release, and "Prediction" could be the probability the release has a negative sentiment. There are m hidden layers ($m = 2$ for illustration purposes, although $m \geq 3$ is more typical for deep learning models). Each hidden layer has n nodes (n is the same in both layers of this illustration but does not have to be equal in practice). In this model, the layers are all fully connected, meaning each node in one layer is connected to all nodes in the preceding and following layers, but this is not true for all models (e.g., CNN). The first hidden layer tends to focus on modeling low-level concepts, such as words, whereas later layers tend to capture higher-level combinations of words, such as concepts and ideas.

Deep learning remained relatively obscure until it was used in 2012 to classify images in an annual contest called ImageNet, where a very large CNN model was considerably more accurate than the traditional machine learning models that had been used until that time (Krizhevsky et al. 2012). The so-called AlexNet model marked a fundamental transition toward deep learning models for image recognition—a breakthrough event referred to as the ImageNet moment.

Even though CNNs were originally created to process images, they were later found to process natural language quite well (LeCun et al. 2015). Unlike images, however, natural language is sequential in nature and would benefit from having a larger context (e.g., other words surrounding a given word) in one dimension, perhaps an entire sentence rather than just a few words. CNNs are limited in capturing the sequential nature of language because they use a limited window for the two-dimensional spatial context. *Recurrent neural networks* (RNNs) were later developed by adding time-aware loops into the ANN architecture to better understand language dependencies and context, allowing the mapping of an input sequence to an output sequence. This mapping makes RNNs particularly well suited for tasks when the order of input tokens has strong meaning, such as language translation, word embeddings, and sentence classifications (Aggarwal 2018).

Due to their looping feature, RNNs tend to have good short-term "memory" in the sense of making decisions based on the immediately surrounding words but have poor long-term memory as the context gets farther away. *Long short-term memory* (LSTM) is an improvement to the RNN looping mechanism by allowing for improved awareness of additional context (Hochreiter and Schmidhuber 1997).[15] Better awareness of long-distance connections among words, coupled with the ability to forget when the context gets farther away or becomes irrelevant, is referred to as "*attention*" in NLP. The addition of LSTM to traditional RNNs allowed for tremendous advances in deep learning.

Later, a groundbreaking study entitled "Attention Is All You Need" (Vaswani et al. 2017) removed the RNN looping mechanism and showed that the attention mechanism was sufficient

15. LSTM also addresses a severe technical challenge to RNNs known as the vanishing gradient problem (i.e., the gradient becomes so small that it prevents the weights from changing in earlier layers, thus stopping the learning). LSTM does not address the exploding gradient problem (i.e., the gradient becomes so large that the loss function oscillates across minima and never settles at the global minimum).

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on its own to produce state-of-the-art model quality. This attention-based model, which the authors call a *transformer*, has come to dominate NLP advances in recent years. It works through varying its attention by emphasizing more relevant and important context clues and minimizing less important context, a technique known as "soft attention." Google's implementation of a transformer, BERT (bidirectional encoder representations from transformers), is now in wide use, including in the Google search engine itself, because of its superiority over other deep learning options (Devlin et al. 2019).[16]

In summary, deep learning models have three advantages over traditional machine learning. First, they can learn complex relationships. Second, they can automate the process of *feature engineering*, including feature extraction and selection. Third, pretrained deep learning models are available so that researchers only need to fine-tune a model for their own settings. This benefit comes from *transfer learning*: the architecture of deep learning that allows the knowledge gained from one task to be applied to a similar one. Fine-tuning a *pretrained model* requires a relatively small amount of training data. Thus, even though training a deep learning model from scratch may require much more data than training a traditional machine learning model, if researchers use a pretrained deep learning model, the amount of training data they need to fine-tune the model is much less than what they would need for traditional machine learning.

2.2 Text representation approaches

There are two approaches to representing text as quantitative inputs to an NLP model. The BOW approach was historically common, whereas the word embedding approach is more recent. In theory, either approach can be used for any NLP model. In practice, because of convention, familiarity, task requirements, and efficiency, BOW is typically used for simple transformations, text comparisons, and traditional machine learning models, whereas word embeddings are often used for deep learning models.

2.2.1 BOW approach

Under this approach, a document is represented by a vector, where each element corresponds to one unique *token* in the *corpus* and contains a value of either one for present or zero for not present (referred to as one-hot encoding). A common variation is for the vector elements to contain a raw count of the number of occurrences of the corresponding token.[17] The approach is appropriate when researchers are interested in specific keywords such as "tax," "non-GAAP," or "allowance" or when researchers intend to compare the use of exact language between two documents. In other situations, the approach may oversimplify much of the richness of human language and cannot represent relationships between the semantic meanings of words. For example, "high" and "low" are opposites, and "nice" and "kind" are synonyms, but the BOW approach ignores these relationships. Moreover, the approach may require input vectors with tens of thousands of dimensions, resulting in an extremely large number of weights to be estimated.

2.2.2 Word embeddings

A word embedding is a text representation approach that maps each word to a feature vector, which consists of the weights for abstract features that collectively capture word usage.[18] As a

16. Closely related BERT precursors, which are less popular, include ULMFiT (universal language model fine-tuning) (Howard and Ruder 2018) and ELMo (embeddings from language model) (Peters et al. 2018).
17. Phrases can be used in place of words, but the dimensionality of the vector grows quickly, making it difficult to work with due to resource constraints. The use of phrases (n-grams, for example, bigrams for two-word phrases and trigrams for three-word phrases) may also lead to a problem known as the curse of dimensionality; that is, the vectors are so sparse (i.e., most elements are zero) that no vectors are "close" to any other vector.
18. The field used to call these feature vectors "word vectors" (Mikolov et al. 2013) but now more often calls them "word embeddings." Even though under BOW a document is represented as a vector of unique words, the phrase "word vector" is reserved for word embeddings.

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result, words with similar meanings have similar vector representations. The word embedding approach has three benefits. First, it reflects, to some extent, the semantic relationships of words and therefore bridges the human understanding of language to that of a machine. Second, it reduces the dimension of inputs from thousands of unique words to hundreds of features and thus increases machine learning efficiency. Last, it allows linear operations of feature vectors to be meaningful. For example, the vector operations for ["king"—"man" + "woman"] would yield a feature vector close to that for "queen."

There have been three generations of word embedding models. Bengio et al. (2001) proposed the idea of word embeddings using a probabilistic approach based on statistical correlations but not context. The second generation of word embedding techniques uses shallow neural networks—ANNs with fewer than three hidden layers—and unsupervised learning to discover word relationships in the training data. For example, researchers at Google developed a popular model known as Word2Vec, which learns how words are used by examining other words in the immediate context, say the surrounding 5–10 words (Mikolov et al. 2013). Another word embedding technique in wide use is GloVe, which was developed at Stanford University (Pennington et al. 2014). These models are typically trained on an extremely large corpus of text, such as Wikipedia, and therefore learn the usual context in which a word is used. Both models are standalone and pretrained; neither is deep learning.

The second-generation models produce vector representations of the words that reflect the training data. For example, if "bank" is used in a financial context in 60% of the corpus, refers to rivers in 30%, and airplanes in 10%, the word embedding will encode this relative frequency. The word "bank" will always be represented in the same way regardless of the context in which it is used in the documents of researchers' interest. Researchers may use a training sample in their own setting to fine-tune these word embedding models to improve text representations for a specific context.

Although Word2Vec and GloVe are still widely used, their inability to reflect the context of the current document is an obvious weakness. This shortcoming is addressed by the third generation of embeddings—an expanded type of embeddings provided by deep learning language models such as ELMo and BERT.[19] ELMo is a two-layer bidirectional language model built on an LSTM architecture and produces word embeddings that consider the entire sentence containing a given word in the current document. BERT, which can also be used as an embedding model, uses soft attention to account for the precise way a word or even a character is used in the current context. As a result, a word with multiple meanings or used in different tenses has different vector representations. For example, the word "bank" in the sentences "they obtained a loan from the bank" and "they met at the river bank" is represented by different feature vectors. The third-generation embedding models can capture both the syntax and semantics of textual documents.

In summary, NLP methods vary by how textual data are represented as inputs to the model and by the type of model used. Although advances have been made to traditional machine learning models, such as gradient boosting tree models and topic modeling, most recent advances in NLP have occurred in deep learning, especially transformers. Although transformers are the state of the art, newer models will almost assuredly transcend them, whether through new architectures or combining the best features of existing architectures. Rapid advances in the field require researchers to stay up to date with high-level trends and developments so as to distinguish between minor improvements and fundamental shifts in NLP.

3. Applications of NLP methods in accounting

We now discuss the applications of textual analysis in accounting research. We begin by grouping textual analysis publications by the NLP task. Studies performing more than one task are placed into multiple categories. Among the 207 publications we have reviewed, 37.2% measure

19. Language models primarily encode the use of language in the form of a quantitative model. One use of the model is to predict what word comes next or more generally assign probability to a piece of text sequence.

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the sentiment of disclosure; 25.6% measure disclosure readability; 18.8% count words, sentences, or articles that contain keywords as a proxy for the quantity of textual information; 10.1% compare disclosures to determine similarities or differences; 6.8% identify forward-looking information; and 2.4% detect disclosure topics (Figure 3, panel G, in Appendix 2). In each subsection below, we discuss the nature of a task, the NLP methods commonly used in accounting, and newer methods that might result in improved measurement of the construct.[20] Note that the discussed applications are intended to be representative but not exhaustive.

3.1 Sentiment

Sentiment analysis is one of the most popular textual analysis tasks in social science. It captures whether the tone/sentiment of text is positive, negative, or neutral. Applications are broad and include the sentiment measurement of management-investor interactions, consumer feedback, political speech, and attitudes of the general population. In accounting, researchers have examined the tone of mandatory disclosures filed with the SEC, earnings press releases, conference calls, media articles, social media posts, analyst reports, and CEO interviews (Henry 2008; Kothari et al. 2009; F. Li 2010b; Feldman et al. 2010; Davis and Tama-Sweet 2012; A. Huang et al. 2014; Davis et al. 2015; Bochkay et al. 2020).

3.1.1 Dictionary method

Sentiment analysis is most often performed using a dictionary method. For example, of the surveyed accounting publications that conduct sentiment analysis, 97.4% use this method.[21] A dictionary is a list of words or rules about the connections between words (e.g., allowing for an extra word between "not" and "agree" in sentences such as "we do not necessarily agree on this point") for identifying features of interest from textual data. The word list and rules determine the lexical items that can reflect the content domain for the underlying construct. The dictionary method for sentiment uses a predetermined dictionary of positive and negative words. Researchers count the number of times positive and negative words occur in a specific document. The sentiment of the document is usually calculated as the difference between positive and negative counts, scaled by total words in the document. The method is straightforward, and one can easily check which individual words contribute to the overall sentiment score.

Despite its simplicity and ease of use, the dictionary method has significant limitations. First, it does not consider context. For instance, "good performance" and "public good" would both be coded as positive because the word "good" is classified as positive in existing single-word sentiment orientation dictionaries. This problem also persists for longer contexts. For example, if the dictionary categorizes "growth" as a positive word, consider two statements: "we expected growth of 10%" and "however, we experienced growth of 5%." The dictionary method would incorrectly classify these two sentences as positive because of the word "growth," ignoring that "however" contrasts the positive expectation with the less positive realization.

Second, the dictionary method assumes that all positive or negative words in a dictionary carry the same weight. For example, the method would assign an equal weight to words "bad" and "terrible" even though "terrible" is generally more negative than "bad." To account for this problem, Bochkay et al. (2020) create a dictionary of words that are ranked on both sentiment (positive vs. negative) and extremity (moderate vs. extreme). Third, researchers need to decide whether and how to account for negators and intensifiers (e.g., "we cannot assure," "not good," and "much worse"), synonyms and related words (e.g., "bad situation" and "disaster"), and idioms ("dodge a bullet"), especially if these terms are not in the input dictionary.

20. We omit disclosure quantity (e.g., document length) because it requires no further explanation.
21. Pang and Lee (2008), Henry and Leone (2016), and Heitmann et al. (2020) provide surveys of sentiment analysis in different disciplines.

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Finally, sentiment dictionaries developed for one domain may not work well for others. For instance, generic dictionaries of positive and negative words (e.g., Harvard General Inquirer) do not work well for measuring the sentiment of business text (Loughran and McDonald 2011). Dictionaries created for measuring sentiment of formal written text, such as SEC filings, may not capture the sentiment of less formal and arguably richer spoken language used in earnings conference calls, investor conferences, IPO roadshows, and CEO interviews (Bochkay et al. 2020).[22]

3.1.2 Machine learning methods

In contrast to the dictionary approach, machine learning can account for peculiarities of language such as negators, intensifiers, and different meanings based on part of speech or context. This approach uses annotated training data (e.g., sentences classified as positive, negative, or neutral) to discover relationships between input text features and output sentiment labels. A trained model is then used to predict the sentiment of unseen text.

Across disciplines, several types of machine learning models have been used for sentiment analysis. Traditional machine learning models include NB (F. Li 2010b; Azimi and Agrawal 2021), SVM (Howard and Ruder 2018), and RF (Frankel et al. 2021). Among deep learning models, CNN, LSTM, and transformers (e.g., BERT) can all perform various text classification tasks, including sentiment analysis (Zhang et al. 2015; Wang et al. 2016; Colón-Ruiz and Segura-Bedmar 2020).

Heitmann et al. (2020) conduct a meta-analysis of 216 publications that report the accuracy of two-class (positive/negative) sentiment analysis on product/service reviews, social media text, and media articles using dictionary, traditional machine learning (NB, SVM, and RF), and deep learning methods (CNN, RNN, and LSTM). The authors observe that the largest leap in accuracy is obtained when switching from the dictionary approach to traditional machine learning, with an average increase in accuracy of 9.13 percentage points, from 73.49% to 82.62%. Switching from traditional machine learning to deep learning gains an average of 4.56 percentage points in accuracy, from 82.62% to 87.18%.

Despite improved accuracy relative to the dictionary approach, only a few accounting studies have used traditional machine learning or deep learning models for sentiment analysis. F. Li (2010b) and X. Huang et al. (2014) use NB; Frankel et al. (2021) use RF. Accounting researchers have only recently started to use transformers. For instance, Siano and Wysocki (2021) demonstrate the use of BERT using a small sample as a proof of concept. The noteworthy development of transformer models for users of financial text is FinBERT, which uses the architecture of BERT but is trained on financial text. There are two mature versions of FinBERT. The version developed by A. Huang et al. (2022) is trained on financial reports, analyst reports, and earnings conference call transcripts. The version developed by Liu et al. (2020) is trained on financial news and chatroom discussions in Reddit.

A. Huang et al. (2022) use human-annotated sentences of financial text of three sentiment classes (positive, neutral, and negative) to fine-tune FinBERT for sentiment analysis. They report that the overall accuracy of FinBERT is 88.2% for their test sample, substantially higher than that of the dictionary approach (62.1%), NB (73.6%), SVM (72.6%), RF (71.9%), CNN (75.1%), and LSTM (76.3%). The accuracy of FinBERT is also higher than that of BERT (85.0%), likely because FinBERT is pretrained on large quantities of financial text. A. Huang et al. (2022) report the sensitivity of model performances to varying sizes of training sample. FinBERT maintains an

22. Over the years, much work has gone into optimizing dictionaries for measuring the sentiment of business text. Henry (2008) and Loughran and McDonald (2011) developed dictionaries of positive and negative words for use in formal business text (e.g., financial reports). Loughran and McDonald's financial sentiment dictionary has gained much popularity. For instance, their dictionary is frequently used to measure the tone of SEC disclosures (Bochkay and Levine 2019), earnings press releases (Davis and Tama-Sweet 2012; X. Huang et al. 2014), conference calls (Allee and DeAngelis 2015; Bochkay et al. 2019), and analyst reports (Twedt and Rees 2012; Bellstam et al. 2021).

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accuracy of at least 80% even when the training sample reduces to only 10% of the original size, whereas the accuracy of traditional machine learning drops below 62%. When 50% of the original training sample is used, the accuracy of FinBERT is 85.7%, followed by BERT (82.8%), LSTM (75.4%), and CNN (72.4%), whereas traditional machine learning performs at below 70%. These comparisons indicate that deep learning gains increased performance at a lower cost than traditional machine learning.

In summary, machine learning methods outperform the dictionary method, with deep learning further outperforming traditional machine learning. The availability of pretrained deep learning models, such as CNN, LSTM, BERT, and FinBERT, has substantially reduced implementation costs. Even though researchers need to create annotated data for supervised machine learning, transfer learning allows for a substantially smaller amount of training data to fine-tune a pretrained deep learning model than what would be required for traditional machine learning models, which typically must be trained from scratch.

3.2 Readability

Readability means the cognitive load on a reader in comprehending text. This load is influenced by many factors, including word complexity, word familiarity, syntax, internal connections between ideas, and the reader's knowledge (Martinc et al. 2021).

3.2.1 Conventional methods

Researchers tend to focus on the lexical and syntactic features of text, such as the number of complex words, length of a sentence, or the length of the entire document, to simplify the measurement of readability. Three popular methods are used in accounting research: (i) document length (F. Li 2008; Lawrence 2013) or document file size (Loughran and McDonald 2014; Y. Li and Zhang 2015), (ii) Gunning Fog Index (FOG; F. Li 2008; Guay et al. 2016), and (iii) the prevalence of "writing faults" such as passive voice and overwriting (Bonsall et al. 2017).

Using document length or file size is the simplest way to measure readability and has an intuitive appeal: longer documents take more time to read, increasing the cognitive load on readers.[23] This method has two limitations. First, lengthy text might serve to better explain a complex issue that would be hard to understand if the text were shorter. Second, longer text may contain more information, making it difficult to disentangle readability from the quantity of information.

The FOG or the Flesch–Kincaid readability score is constructed based on the number of words per sentence and the percentage of complex words (i.e., multisyllable words) in text. Intuitively, longer sentences and more complex words increase the cognitive load on readers. The advantages of this method are simplicity and ease of calculation. The method has two disadvantages, however. First, it has been criticized for oversimplification (Crossley et al. 2017). Second, some complex words are not necessarily difficult for readers in a given domain. For example, "individual," "financial," "company," "television," "advertising," and "marketing" all contain more than two syllables and are therefore classified as complex, but they are commonly used and easy to understand in a business context.

Subsequent studies attempt to address the above problems. For example, Bonsall et al. (2017) propose the Bog Index (BOG), which considers the average sentence length in the document, word complexity (e.g., heavy words, abbreviations, and specialist terms), word familiarity, and writing faults. The authors demonstrate that BOG outperforms commonly used readability measures in capturing the readability of financial text. BOG, however, requires a commercially available add-on to Microsoft Word.

23. Note that "file size" is a noisy proxy if a document contains images, tables, figures, or symbols. Bonsall et al. (2017) caution that this measure might induce bias.

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3.2.2 *Machine learning methods*

Several studies use supervised machine learning to address the shortcomings of the conventional methods. These studies use readability scores assigned by human annotators to train models to predict the readability of unseen text. For example, Petersen and Ostendorf (2009), Vajjala and Meurers (2012), and Madrazo Azpiazu and Pera (2020) use traditional machine learning models to characterize text as a set of readability features that include (i) conventional features, such as FOG; (ii) discourse features that measure text coherence and cohesion; and (iii) features such as how commonly used a word is. Filighera et al. (2019) demonstrate high performance of ANN-based word embeddings to measure readability.

A main disadvantage of the above methods is that the measures may not generalize to different contexts (e.g., Wikipedia text vs. SEC filings vs. earnings conference calls) and different audiences (e.g., high school students vs. doctoral students and sophisticated vs. unsophisticated investors). To overcome this problem, Martinc et al. (2021) propose an unsupervised method based on deep learning. They model word sequence in a sentence and assign a readability score for the sentence based on the predictability of the words it contains. If a given word in a sentence is correctly (not) predicted by the model based on word location and preceding words, then the text is more (less) readable. The authors demonstrate that their readability score has more consistent performance across different corpora (e.g., textbooks vs. news articles) than conventional readability measures.

3.3 *Disclosure similarity*

Another popular textual analysis task is measuring the similarities or differences between disclosures. Two documents are considered similar if they contain similar words or imply similar meaning. Applications include identifying plagiarized versus unique content, measuring overlap of products or services, capturing changes in communication over time, and documenting clustering.[24]

3.3.1 *BOW method*

Cosine similarity is one of the most common ways to measure disclosure similarity (Schütze et al. 2008). The conventional method for calculating cosine similarity is representing each document by BOW. Values closer to one indicate a high degree of similarity, whereas those closer to zero indicate low similarity. The use of raw word counts is referred to as term frequency (tf) weighting. An alternative weighting scheme is term frequency–inverse document frequency (tf–idf), which downweights words that appear in many documents in the corpus and therefore incorporates corpus-based information (Salton and Buckley 1988).

The BOW-based cosine similarity method is intuitive. For example, the sentences "revenues increased by ten percent" and "we had an increase in revenues of ten percent" will be ranked as very similar because they contain the same words for the most part. The method excels in situations where the exact word choice is important, such as plagiarism detection and year-over-year modifications to a document. The method, however, treats all unique words as independent inputs and does not consider semantic relations between words. For example, the sentences "revenues increased by ten percent" and "sales went up by ten percent this period" will be ranked as less similar because the wording differs even though they deliver the same message.

Several accounting studies examine the similarities or differences of corporate disclosures using BOW-based cosine similarity. S. V. Brown and Tucker (2011) track a firm's changes in the management discussion and analysis (MD&A) section of 10-K reports from one year to the next as a proxy for the amount of new information in the current year's MD&A. Peterson et al. (2015)

24. For example, Hanley and Hoberg (2010) use textual similarity to detect unique content of IPO firms' prospectuses. Hoberg and Phillips (2016) use similarity of business descriptions in 10-K filings to generate sets of industry peers.

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examine a firm's year-over-year similarity in its accounting policy notes that accompany the financial statements as a measure of accounting consistency. S. V. Brown and Knechel (2016) compare textual similarity of annual reports across firms to determine whether disclosures by firms with the same auditor are more similar than disclosures by firms with different auditors. S. V. Brown et al. (2018) use BOW-based cosine similarity to examine the spillover effect of SEC scrutiny of corporate risk factor disclosure.

3.3.2 Word embedding method

Combining cosine similarity with word embeddings can capture semantic similarity. Kusner et al. (2015) use Word2Vec word embeddings as inputs to the cosine similarity model to account for synonyms and related words. The authors compare their model with seven baseline models and report that their model produces the highest accuracy of document classification. Gaulin and Peng (2021) use the Doc2Vec word embedding model, which extends Word2Vec by adding document vectors to generate a vector representation for each document. The authors use these vectors as inputs to the cosine similarity model to capture semantic similarity of executive compensation disclosures across firms.

In summary, the BOW-based cosine similarity method is appropriate when researchers are interested in comparing precise words chosen, whereas the word-embedding cosine similarity method can capture semantic similarity. Researchers should choose between these two methods based on which aspect of document similarity they aim to capture.

3.4 Forward-looking statements

Temporal orientation of sentences (i.e., past, present, or future) is a classification task. Forward-looking statement classifications have received much attention in accounting.

3.4.1 Dictionary method

The conventional method for classifying forward-looking statements is using a dictionary, where researchers supply a list of words or phrases that are likely future-oriented. F. Li (2010b) uses a list of future-oriented verbs (e.g., "will," "plan," and "expect") to determine whether a sentence is forward-looking. Muslu et al. (2015) improve this list by using bigrams (e.g., "next year" and "coming year"), selected conjugations of future-oriented verbs (e.g., "we project" and "company expects"), and references to future dates. Bozanic et al. (2018) further improve the dictionary by eliminating phrases likely used in cautionary language (e.g., "coming period" and "coming year").

The dictionary method is simple and intuitive but sacrifices accuracy. Even though the method has a low Type II error rate (i.e., it can effectively find all statements that are truly forward-looking), it results in a high Type I error rate (S. V. Brown et al. 2021). That is, it may misidentify a substantial percentage of statements that are not forward-looking as forward-looking. Most sentences in a financial document are not forward-looking and therefore vulnerable to this problem.

3.4.2 Machine learning methods

Forward-looking statement classification can be done with a variety of supervised machine learning models. S. V. Brown et al. (2021) fine-tune a pretrained CNN model by providing it with human-annotated training data consisting of (i) specific future-oriented sentences; (ii) generic future-oriented sentences (e.g., risk disclosure or cautionary language); and (iii) non-future-oriented sentences. They label the first type as forward-looking and the other types as not forward-looking. Their algorithm correctly identifies a sentence as forward-looking versus not forward-looking 88.2% of the time. In contrast, the dictionary method with the best available dictionary of Bozanic et al. (2018) results in an accuracy rate of 73.4%. The improvement by deep learning is attributable to its ability to detect subtle relationships in textual data. For example, it

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would be rather difficult for the dictionary method to distinguish between generic versus specific future-oriented sentences. In contrast, deep learning can mimic human annotators in distinguishing between word patterns in these two types of sentences.

3.5 Topic discovery

Topic discovery is designed to summarize a document by grouping words into themes called topics. When a document contains a topic such as "production cost," one would expect to see words such as "production," "efficiency," "materials," "labor," and "overhead" more frequently. On the other hand, the "advertising" topic would be associated with more frequent use of words such as "customer," "satisfaction," and "marketing." A topic model discovers a set of latent topics and determines their distribution in a given document. For example, a document may be 70% about production cost and 30% about advertising.

The most common topic model is LDA, which was originally designed to be used with individual word counts under the BOW text representation approach. Campbell et al. (2014) first used LDA in accounting, even though their application is unconventional. The authors use the model to identify words that occur frequently in a firm's risk factor disclosure section of a 10-K report and add these words to their initial risk disclosure keyword list for measuring the quantity of the firm's risk disclosure. A. Huang et al. (2018) use LDA to compare the content of earnings calls and subsequent analyst reports. Dyer et al. (2017) use LDA to examine the temporal evolution of disclosure topics in 10-K reports. N. C. Brown et al. (2020) use LDA to supplement traditional numerical fraud prediction models with textual information.[25] We discuss recent advances in topic modeling in section 6.2.

4. Construct validity of text-based measures

As panel F of Figure 3 (Appendix 2) shows, accounting researchers typically use textual analysis to create a measure to proxy for an underlying construct—that is, a concept or characteristic that cannot be directly observed but can be measured by observing indicators that are associated with it. It is critical for researchers to articulate the construct of their interest and then establish the construct validity of their measures, which is the ability of an operationalized measure to capture the underlying construct.[26]

Construct validity can be evaluated using two approaches.[27] Under the translation validity approach, researchers evaluate whether the operationalization is a good reflection of the construct. They can assess face validity (i.e., on its face, whether a measure seems to be a good translation of the construct) and content validity (i.e., whether the measure captures the meaning of the construct). The latter evaluation requires directly comparing the measure with the construct. Under the criterion-related validity approach, researchers evaluate whether the operationalization behaves as it should, given researchers' theory about the construct. Evaluations are indirect and based on how the operationalization performs using one of four criteria: convergent validity, concurrent validity, predictive validity, and discriminant validity. We explain these concepts and provide examples in Appendix 3. Depending on the NLP method and data availability, researchers might be able to evaluate only some of the facets of construct validity. If feasible, evaluating content validity is preferred. In the next three subsections, we recommend validation procedures for

25. In finance, Hanley and Hoberg (2019) use LDA to detect emerging risks in the financial sector and Bellstam et al. (2021) use LDA to measure innovation among S&P 500 firms.
26. Construct validity is central to establishing internal and external validity and therefore the overall validity of a study. Both internal validity and external validity are concepts relevant to examining causal relationships. Because textual analysis itself typically does not test a causal relationship, we focus on construct validity, not internal or external validity.
27. See Bloomfield et al. (2016) and *Research Methods Knowledge Base*, an e-book written by Professor William M. K. Trochim and hosted at Conjointly (https://conjointly.com/kb/measurement-validity-types/).

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text-based measures created by dictionary methods, supervised machine learning models, and unsupervised machine learning models.

4.1 Construct validity of measures created by dictionary methods

Researchers should validate their dictionary; doing so will strengthen content validity. It would be ideal if an established dictionary already exists for the researcher's setting. For example, Loughran and McDonald (2011) provide a dictionary for "uncertainty" in financial contexts. Most often, researchers must build their own dictionary or adapt one to their setting. We use an example to describe the procedure that researchers can use to validate a new dictionary. Carrizosa and Ryan (2017) identify loan contracts that contain a covenant requiring the borrower to regularly provide prospective financial statements to the lender. The authors manually examine the covenant section of 110 loan contracts randomly drawn from their sample contracts and identify the specific words and context where the covenant of interest appears. Using these words and context as their dictionary, they wrote Python code to automatically identify contracts that contain this covenant. They then compare this identification with human-coded labels in another randomly selected subsample and adjust the dictionary until achieving desired accuracy.[28]

We advise that researchers check the content validity of their measure by using third-party human raters. The raters can be research assistants who are blind to the research objective as well as to the sample composition if the sample is stratified. The raters can also be subjects in an experiment, which aims at comparing human versus algorithmic coding. For example, Bonsall et al. (2017) hire MTurk workers to rate the readability of letters to shareholders and manipulate the instrument so that some workers receive more-readable letters while others receive less-readable ones. The authors correlate their proposed readability measure with the workers' readability ratings. Muslu et al. (2015) ask MBA students to identify forward-looking sentences in 50 randomly selected MD&A disclosures and compare those identifications with ones by an algorithm, demonstrating the content validity of their measure.

4.2 Construct validity of measures created by supervised machine learning

Supervised machine learning has a built-in mechanism for validation. The annotated training data set is typically split into the training, validation, and testing samples. The *training sample* is used to fit the model and determine the weights, which represent data relationships. The *validation sample* is a hold-out sample for assessing the ability of a fitted model to predict out-of-sample observations. Researchers adjust the model hyperparameters and use the training and validation samples iteratively until the model achieves desired performance. The *testing sample* is another hold-out sample to assess model performance a single time after all training is complete. With this model construction and testing process, content validity is assessed when researchers use the final fitted model on unseen data in the testing sample and compare their measure with the annotated labels.

Several performance metrics are available to quantify the above comparisons. *Precision* is the number of true positives divided by the number of items that the model identifies as positives (e.g., of all the forward-looking sentences identified by the model, how many are correctly identified?). *Recall* is the percentage of positive items in the sample that were correctly identified as such (e.g., of all the forward-looking sentences in the document, how many are identified as such by the model?) and is one minus Type II error. The *F-score* is a weighted combination of precision and recall, where the researcher chooses the relative weight for each metric. Many studies

28. Fine-tuning a dictionary or building a new one can be an iterative process. This process resembles the automated training process in machine learning, except that here researchers iteratively adjust the rules manually and evaluate the fit. Researchers may use a third party's help to build a dictionary. For example, Bochkay et al. (2020) hire MTurk workers to rate a set of 23,355 keywords on a scale from "extremely negative" to "extremely positive."

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referring to the "*F*-score" actually use the F_1-score, which is the harmonic mean of precision and recall with equal weight.

Precision, recall, and F_1-score ignore true negatives—the negatives that have been correctly identified, for example, non-forward-looking sentences that have been correctly classified—unless researchers use the mean value of the metrics reported for all classifications (e.g., both forward-looking and non-forward-looking sentences).[29] Many researchers use accuracy (i.e., how many predictions are correct?) as a performance metric. Heitmann et al. (2020) report that accuracy is the most common performance metric in sentiment analysis. Accuracy takes true negatives into account but gives equal weight to all observations. If the researcher is particularly interested in true positives but the sample is comprised of largely negatives, then the accuracy metric might be inappropriate. Other performance metrics are Type I and Type II errors and the area under the curve, known as ROC-AUC. As in traditional statistical analysis where researchers must balance Type I and Type II errors, it is up to researchers to decide the relative importance of true positives versus true negatives in evaluating their text-based measures.

In addition to train-valid-test validation, *K*-fold cross validation is commonly used for traditional machine learning. The training data are randomly partitioned into K equal parts. Then K experiments are performed, where each time one part is saved for out-of-sample testing and the remaining $K - 1$ parts are used for training. The performance results are averaged across the K experiments. This technique is especially useful for small samples because all training observations are used at some point for the model to learn. Another benefit of *K*-fold cross validation is that it can show whether a model is sensitive to the randomness of the training sample. For example, F. Li (2010b) reports stable model performance when K varies from 3 to 50. This validation procedure is uncommon for deep learning mainly because it is computationally costly. In deep learning, two alternative validation procedures are "dropout" and "batch training." In dropout, some nodes are randomly dropped out of the neural network, with the goal of reducing overfitting. In batch training, batches (i.e., subsamples of the observations) are formed randomly from the training sample, with the goal of reducing sensitivity to the composition of the training sample.

Because supervised machine learning starts with an annotated sample, the quality of *annotation* is extremely important for the content validity of the measure produced. Otherwise, garbage in results in garbage out (Geiger et al. 2021). Annotators can be researchers, their research assistants, or online third parties. Annotation might be the most labor-intensive part of developing a supervised machine learning algorithm. However, the availability of online platforms (e.g., MTurk, Brat, CrowdFlower, and WebAnno) and gig workers has made it convenient and affordable for researchers to recruit third-party annotators. In addition to freeing up researchers' time, with proper screening these resources may better represent the intended receivers of corporate disclosures than researchers themselves.

We advise using at least two human annotators so that the consistency of annotations can be assessed. A rule of thumb for the agreement between two annotators for binary classifications is 80%. This rate corresponds to the minimum value of Kappa statistic for "substantial" success regions in Landis and Koch (1977) and is the rule of thumb used in most experimental research.[30]

29. If researchers are interested in identifying forward-looking sentences, the actual forward-looking sentences identified as such are called "true positives" (TP) and those failed to be identified are called "false negatives" (FN); the actual non-forward-looking sentences identified as such are called "true negatives" (TN) and those failed to be identified are called "false positives" (FP). Precision is defined as TP/(TP + FP), recall is TP/(TP + FN), and accuracy is (TP + TN)/(TP + FP + TN + FN). Type I error is FP/(TN + FP), and Type II error is FN/(TP + FN).

30. Landis and Koch (1977) provide Kappa statistic values for "poor," "slight," "fair," "moderate," "substantial," and "almost perfect" success regions. Because the baseline agreement rate for two independent coders in binary classifications is 50%, the minimum values of Kappa statistic for "moderate," "substantial," and "almost perfect" correspond to the agreement rates of 70%, 80%, and 90%, respectively. Note that a lower acceptable agreement rate is expected for multiclass annotations.

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Disagreements among annotators can inform researchers of the cases that would be challenging for the algorithm to differentiate. If it is difficult for humans, it is likely difficult for the machine. Accordingly, researchers can adjust their training objective or provide more training observations for the challenging cases.

Instead of human annotations, some studies label their output variable in the training data using a variable collected from financial data. We refer to this method as "indirect annotation." For example, Siano and Wysocki (2021) and Frankel et al. (2021) use indirect annotations to train the machine to recognize the sentiment of an entire document. Siano and Wysocki (2021) annotate a quarterly earnings announcement press release as positive (negative) if the firm's percentage change in that quarter's revenue from the same quarter in the previous year is above (below) the sample median. The authors acknowledge that accurate annotations often require human intervention, which is costly. Frankel et al. (2021) echo that human annotations are precise, detailed, and tailored to researchers' settings but argue that human annotations are subjective and difficult for others to replicate and may lead to small samples and therefore ineffective training. Frankel et al. (2021) use a firm's stock return on the 10-K filing date to label their training data. Indirect annotation might reduce construct validity because of an extra layer of uncertainty. For example, it is uncertain whether managers use text to express the same sentiment as the label based on revenue changes. If so, the textual information would be redundant. There are also cases when financial performance is favorable, but sentiment is relatively negative. Then, indirect annotation would result in incorrect classification. Text provides context for interpreting financial information; that context might be lost with indirect annotation. Using stock returns to classify a disclosure assumes that stock price fully impounds the disclosed information of interest and does not systematically reflect other information at the event. Thus, indirect annotations might increase the chance that a text-based measure captures something other than the underlying construct.

4.3 Construct validity of measures created by unsupervised machine learning

Unsupervised machine learning is typically used to discover hidden patterns in unlabeled data and produce outputs as data groupings or clusters (e.g., topics), not a single measure. Construct validity then means ascertaining that those groupings represent the concepts of interest to the researcher. Evaluating construct validity is more challenging because of multidimensional outputs.

Researchers can use three procedures to establish the construct validity of measures produced by unsupervised learning. First, to enhance face validity, the researcher can inspect the most common words or phrases associated with a cluster and check whether they appear to have any semantic coherence. For example, A. Huang et al. (2018) read the high-probability words in key topics produced by LDA and their respective sentences to determine semantic coherence. Second, to enhance convergent validity—whether the operationalization is closely correlated with other operationalizations that have been claimed to capture similar constructs—the researcher can select certain clusters and examine whether they vary in periods containing important economic events expected to affect these clusters or correlate the clusters with variables capturing similar constructs. For example, A. Huang et al. (2018) plot the variation of key topics discussed by banks over time and report meaningful changes around the 2008–2009 financial crisis. N. C. Brown et al. (2020) correlate the risk topics identified by LDA in 10-K filings with investors' risk perceptions proxied by future return volatility. Last, to enhance discriminant validity (i.e., whether the operationalization has a low correlation with other operationalizations that capture quite different constructs), researchers can ask human coders to perform "word intrusion" checks. That is, coders are provided with a list of words for each cluster (one word has a low probability for a given cluster but a high probability for a different cluster, while the remaining words have high probability for the given cluster) and asked to identify the low-probability word. Dyer et al. (2017) and N. C. Brown et al. (2020) use this validation procedure.

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Researchers could also ask human coders to identify clusters in a small random selection of documents and then compare the results with the outputs of unsupervised learning. Although this procedure will enhance content validity, it can be labor-intensive because the number of outputs is in the dozens or hundreds. A. Huang et al. (2018) use this validation procedure.

5. Implementation guidance

In this section, we discuss textual data acquisition and preprocessing, model selection, and key decisions in implementing machine learning methods, followed by recommended coding and data resources. For those interested in dictionary-based methods, we provide an implementation checklist in panel B of Appendix 4 and refer readers to Loughran and McDonald (2016, section 6) for details.

5.1 Acquisition and preprocessing of textual data

Textual documents contain a large variety of structure and content. To acquire relevant documents, researchers often must write their own web-scraping code tailored to a specific website and data format. Researchers should ensure that the raw data are obtained in accordance with relevant laws and restrictions.[31] The content of textual documents tends to be inconsistently formatted, for example, plain text, HTML web pages, or PDF documents. A document may contain everything from HTML tags and special symbols to images, tables, and the text itself. PDF documents contain a markup language that is quite different from HTML, requiring specialized libraries to open and parse the content.[32]

Prior to undertaking NLP, researchers must preprocess the text to make it machine usable. Sometimes these procedures may take more time than the formal textual analysis. The text cleaning process generally begins with the removal of HTML tags and non-text characters, followed by the conversion of text into smaller text units (e.g., single words, word bigrams, or sentences). After this step, researchers can remove documents that do not meet the minimum length criteria, for example, documents incorporated by reference, one-word tweets, and empty files. Next, researchers can check for spelling mistakes and ensure proper encoding of special symbols (e.g., apostrophes and hyphens). These steps help increase the quality of input textual data. Working with individual words may lead to the problem of large vocabularies, that is, thousands of unique words in the corpus. To mitigate this problem, researchers often remove stop words (e.g., "a," "the," and "in") because they have little meaning. Furthermore, researchers often *lemmatize* or *stem* individual words to convert inflected or derived words to their base or root form (e.g., converting "increases" to "increase" and "agreement" to "agree"). Panel A of Appendix 4 summarizes typical data acquisition and preprocessing procedures.[33]

However, stop words and inflected/derived words can be meaningful sometimes. For instance, if researchers are interested in determining the tense used or capturing the writing style, preserving the original text structure is important. In other cases, stop-word removal and lemmatization or stemming are not necessary because the selected NLP method has built-in mechanisms to account for frequent words and inflected or derived words. For example, these procedures are not needed if word embeddings are used as inputs to the NLP model. There is also

31. At a minimum, researchers should follow the law (e.g., the Digital Millennium Copyright Act), the terms of service provided by the subscribed data source, and any other relevant restrictions for web scraping. For non-subscribed websites, the convention is to retrieve a file called "robots.txt" from the website and follow the rules defined in that file or posted elsewhere on that website.
32. Some PDF files, especially older ones, do not contain machine-readable text. Each page of the document is effectively a picture of what the text would look like if it were printed. Optical character recognition techniques are needed to convert the document to a machine-readable format. A substantially larger file size than normal strongly suggests this type of PDF. Fortunately, such PDF files have become uncommon in recent years.
33. There are packages available in Python and R that facilitate cleaning so that researchers do not have to code every step themselves.

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no need to remove stop words when tf–idf weighting is used. Deep learning models usually do not benefit from lemmatization, stemming, or stop words removal. In fact, stop words can provide semantic context to deep learning models. Moreover, BERT can process subwords (e.g., "vol," "ati," and "lity") instead of words.

5.2 Model selection

To extract useful information from text, researchers must decide which model to choose from various NLP models, which range from simple transformations and text comparisons to traditional machine learning and deep learning.[34] We recommend five factors to consider: functionality, simplicity, familiarity, model performance, and computing costs. "Functionality" means whether a model serves the intended function. For example, if a project requires precise similarity of words in disclosures, the BOW cosine similarity model is the best, and machine learning models are inapplicable.

"Simplicity" means that if multiple models can serve the purpose, the simplest is preferred. More complex models should be chosen only if something is gained for the effort invested. For example, if a researcher wishes to control for readability in a relatively less important analysis, FOG might be a good choice. However, if readability is a primary focus of the study, the benefits of machine learning methods may justify the additional effort. If a project requires finding companies in textual documents that use "blockchain" or its variations, a simple transformation model with a list of keywords is optimal.

A researcher may already be familiar with various traditional machine learning methods. Some of these models, such as NB and SVM, have been used in different contexts (e.g., fraud detection, bankruptcy prediction, and sentiment measurement) and proven viable. Others, such as lasso and RF, have recently grown in popularity. For researchers familiar with lasso and RF, the two methods are good choices for modeling linear and nonlinear relationships, respectively. Moreover, there have been advances in traditional machine learning. For example, gradient boosting allows the first tree to capture the first-order economic factor and subsequent trees to focus on what remains unexplained by the previous trees (Bertomeu et al. 2021). Researchers who are familiar with a traditional machine learning model can more easily learn its improvements. Familiarity will minimize effort, help diagnosis, and facilitate implementation.

For researchers unfamiliar with traditional machine learning models, deep learning is a good place to start for three reasons. First, it is a set of variations around the same central model structure (ANN) and thus easier to grasp as a whole, whereas traditional machine learning is many disparate model structures. Second, deep learning typically performs better than traditional machine learning on textual data. Vasyl Harasymiv, senior data scientist at Grubhub, concluded after the 2015 Kaggle Competition that extreme gradient boosting models were the best for structured data (tabular data), whereas deep learning models were the best for unstructured data (text, sound, or visuals).[35] That conclusion was made before the development of BERT. Nowadays, some forms of an attention-based model, such as LSTM and transformers, produce the vast majority of state-of-the-art results in NLP.[36] Bengio et al. (2021, 61) refer to them as the "dominant architecture in many applications," saying "transformers have produced dramatic performance improvements that have revolutionized NLP, and they are now being used routinely in industry."[37] Last, even though pretrained libraries from traditional machine learning exist in special cases, such as TextBlob and VADER, pretrained models exist for all deep learning methods, thanks to the model structure

34. Multiple models can be combined into a single model via *ensemble learning* (Manning and Schütze 1999; Aggarwal 2018).
35. See https://www.kdnuggets.com/2015/12/harasymiv-lessons-kaggle-machine-learning.html
36. See https://github.com/sebastianruder/NLP-progress
37. Yoshua Bengio, Yann LeCun, and Geoffrey Hinton are recipients of the 2018 ACM Turing Award for their work on neural networks and have been foundational figures in deep learning over the past two decades.

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being conducive to transfer learning. Pretrained models are a great convenience to researchers, especially beginners.

Computing costs are a relevant factor in choosing a model. It requires substantially more computing power to train a deep learning model than a traditional machine learning model. We expect, though, the choice for a typical accounting researcher to be between a pretrained deep learning model and a traditional machine learning model to train from scratch. The memory requirements for pretrained models are often modest and fine-tuning is less computationally intensive. The memory requirements for topic models and one-to-many text comparisons could be heavy, though. Among deep learning models, transformers take longer processing time than CNNs. Transformers are also limited in their maximum length (e.g., BERT is limited to processing 512 words at a time), but this is rarely a problem for sentence-level classification because a sentence will rarely contain more than 500 words.

5.3 Implementation decisions for supervised machine learning

Researchers must decide early in a project whether they will use supervised or unsupervised learning and whether their output variable of interest is continuous or categorical. In this subsection, we discuss key decisions for supervised learning and summarize them in panel C of Appendix 4.

5.3.1 Annotations

For supervised learning, researchers provide training data for the machine to learn data relationships; the outputs of these observations have been annotated (labeled). There are two primary types of discrete annotations: *classification* and *named entity recognition* (NER). Classifications are generally simpler to implement in terms of the technology involved, although not necessarily in their degree of human involvement. Human annotation is typically more difficult for a longer text (e.g., an entire article) than a shorter one (e.g., a sentence or paragraph). A typical classification annotation could be formatted in a spreadsheet with one column containing a list of sentences and another column the human-rated categories. Named entities are one or more words in a document that collectively refer to a real-world entity, such as a person, location, or organization.[38] The machine learns to recognize NERs of interest based on items labeled by humans. The machine learns to recognize not only the exact items provided but also how to generalize the process so that new named entities can be discovered even though that exact combination of words was not provided to the machine as part of the training data. Even though the annotation process can be outsourced to third parties, it is helpful for researchers to gain direct familiarity with it. See section 4.2 for best practices of text annotations.

5.3.2 Training data size and sample splits

Due to the high cost of human-annotated data, a key decision for supervised learning models is the optimal quantity of annotated data. If too much data are obtained, costs are higher than necessary. On the other hand, empirical power may be low if the data are too sparse. Anecdotally, simple tasks, such as those that could be performed by keyword searches through regular expressions, may only require a few hundred observations in total. More complex tasks, such as determining the sentiment of a sentence in the context of a full paragraph, may require several thousand annotated cases to fine-tune a deep learning model. If more than 5,000 observations are needed, it is likely that the underlying task needs simplification. If a task is too complex for a human, it will also be too complex for the machine because the model is simply encoding the

38. Many pretrained NLP models already have NER capabilities, including the ability to recognize geographic locations. However, anecdotal experience is that these models often have a high preponderance of both Type I and Type II errors at least for corporate filings. So, these "out of the box" models may be insufficiently accurate and would need fine-tuning.

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knowledge provided to it. Many models should achieve reasonable performance somewhere between these extremes.

When selecting data for annotation, the most important factor is that the data set has enough observations representing each type of potential pattern of interest; for example, sentences with positive, negative, and neutral sentiment. A small sample of randomly selected sentences is unlikely to lead to a well-trained model unless the underlying data are well balanced regarding the patterns of interest. A stratified sampling approach should be used for unbalanced data, where each stratum adequately represents patterns of interest. For example, a sentiment classifier could contain sentences selected using the following strata: (i) words from the Loughran-McDonald (LM) positive list; (ii) words from the LM negative list; (iii) words from the General Inquirer (GI) positive list that are not in the LM positive list to capture words that have generally positive sentiment but not necessarily in a financial context; (iv) same as (iii) but for negative words; and (v) a generous number of randomly selected sentences. Such an approach will lead to a mix of sentences that are positive, negative, and neutral, with an adequate representation of each. This approach includes some borderline cases (e.g., a word is on the GI list but not on the LM list), giving the model an opportunity to learn to distinguish subtle relations.

Researchers can start by annotating a small set of exploratory data to get a sense of the underlying text before gathering the entire stratified sample. The next step is to annotate a small number of observations from each stratum (for a total combined number possibly in the low hundreds). The model is calibrated using the pooled annotated data and performance is assessed using the metrics described in section 4.2. Another set of annotations is then completed, and the model is fit again. The researcher can stop annotating new data once the desired model performance is obtained.

Training data are split into the training, validation, and testing subsamples. A ratio of 2:1:1 is commonly used, but this may lead to much larger validation and testing samples than necessary, especially with large annotation samples (Aggarwal 2018, 222). In small annotation samples, a testing subsample may not be used at all. After assessing performance, some researchers use all annotated examples to train a separate final model to maximize training data, while others prefer a model built using only the training subsample.[39]

5.4 Recommended resources

In panel D of Appendix 4, we provide a list of recommended resources for researchers to get started for their own NLP projects. For instance, Anand et al. (2020) provide extensive Python coding guidance for textual data acquisition, document preprocessing, and information extraction. A. Huang et al. (2022) summarize steps for using FinBERT. Many researchers share code, measures, and implementation guidance on their websites or GitHub (or similar). Many textual analysis papers now include web links to the measures that they have developed or to resources that have facilitated their measure development. We acknowledge that our list is incomplete and is subject to update.

6. Future opportunities

The accounting literature has emphasized the dictionary approach to measuring sentiment, forward-looking focus, and other constructs. Newer NLP methods can improve the measurement of existing constructs, develop new text-based measures, and help answer new research questions.

6.1 New research areas facilitated by deep learning

Deep learning offers a wide range of possibilities. Unsupervised deep learning may learn patterns that were never seen by the researcher or discover concepts that were never known to the

39. The discussion of training, validation, and testing samples may be inconsistent across studies. For example, some studies use "train-test split" to mean the training and validation samples.

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researcher. Supervised deep learning can take a relatively small hand-coded sample, learn to replicate what human coders are doing, and scale it up to a much larger sample of text. This approach applies to any linguistic construct that can be coded by a human, such as specificity, complexity, understandability, concreteness, vagueness, avoidant language, and attribution to an underlying cause. Below we discuss examples of tasks for which supervised deep learning can prove helpful.

Operating and geographic segments are often used by researchers to proxy for constructs such as business and geographic complexity. However, not all companies report segments in the same way or with the same level of granularity. Textual analysis can provide another way of quantifying the diversity of a company's operations or locations. For example, NERs developed through deep learning can identify locations and more precisely label them as distribution centers, satellite offices, stores, competitor locations, etc.

Counting words related to "competition" in regulatory filings is frequently used as a proxy for the level of perceived competition (F. Li et al. 2013). More precision can be obtained by using NERs to analyze companies in the same industry that were specifically discussed by a disclosing company. Deep learning can help further distinguish different types of competition, such as competition in the labor, product, or supplier markets. In a similar vein, deep learning can be used to label corporate risks as company-specific, industry-specific, or economy-wide. Alternately, risks could be classified into other categories, such as competition, climate change, and supply chain interruptions.

Deep learning can also help extract data. For example, XBRL data has been available for approximately 10 years. That data can be used to train a model to extract similar data from the narrative text for companies before the XBRL requirement or that otherwise do not disclose a given item in XBRL. A similar technique can be used to extract risk factors, that is, using Item 1A to train a model that will extract risk factor disclosures in other parts of the 10-K or from an annual report before Item 1A was mandated. Another application is to train a model to recognize a small number of liquidity discussions in the MD&A so that the model can automatically extract liquidity discussions for the full sample.

6.2 Topic analysis of corporate disclosures

Mandatory and voluntary corporate disclosures have long been of keen interest to accounting researchers. Topic analysis of accounting disclosures, such as Bao and Datta (2014) and Dyer et al. (2017), can provide new insights on corporate disclosure decisions and their impact on capital market participants. Currently, LDA is the most popular NLP model for topic analysis, but it has three limitations. First, it uses BOW as inputs, resulting in large vocabularies and high computing costs. Second, the model often produces topics not structured in a way helpful to researchers, and the discovery of specific topics is not guaranteed because the model is unsupervised. This is a problem when researchers are interested in specific topics such as non-GAAP disclosures, tax disclosures, and R&D. For example, the topic model might merge pharmaceutical R&D with other pharmaceutical-specific discussions due to the prevalence of R&D in that industry, thus preventing R&D from being identified as a broader topic distinct from the industries in which R&D occurs. Last, topics produced by LDA may be difficult to interpret or label.

Accounting researchers could exploit recent advances in topic modeling. To overcome the problem of high dimensionality, Dieng et al. (2020) develop the embedded topic model (ETM), which uses word embeddings as inputs to LDA. Word embeddings (e.g., Word2Vec) not only reduce high dimensions, especially rare words, but also represent semantic meanings of words. The authors report that ETM outperforms LDA in terms of topic quality and predictive performance. Similar topic discovery can be accomplished with Doc2Vec and Topic2Vec. The structural topic model allows the inclusion of covariates that serve as a kind of "fixed effects" so that topics can be correlated with non-textual information, such as company, date, and data source (Roberts et al. 2013).

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Another method is seeding the initial topics with words that are pre-grouped into topics of interest, instead of initializing the topics randomly as is typically done (Jagarlamudi et al. 2012). However, there is still no guarantee that the pre-grouped topics of interest will remain in the final model. Researchers can also constrain the topic model to discover only the topics corresponding to the label set provided by researchers; this method is referred to as Labeled LDA (Ramage et al. 2009). If the goal of topic discovery is predicting an outcome variable (e.g., fraud occurrence, bankruptcy, or credit risk), then supervised LDA ("sLDA") can be applied (Blei and McAuliffe 2007), where the weight for each topic is determined by the machine based on training data.

Donovan et al. (2021) use sLDA to discover topics and their weights in conference calls associated with credit risk, which is proxied by credit default swap (CDS) spreads, and then predict the credit risk out of sample for firms without CDS. Two finance studies have introduced techniques that help interpret topics and avoid manual labeling of topics. Hanley and Hoberg (2019) use a word embedding technique (the skip-gram model of Word2Vec, referred to as "semantic vector analysis" in their study), which can predict the context surrounding a given word, to convert the output from LDA into a set of interpretable risk factor topics. Lowry et al. (2020) combine LDA with a technique from information theory called Kullback-Leibler (KL) divergence to link topics in the SEC comment letters to topics in prospectuses and automatically label topics using paragraph headings in the comment letters. The KL divergence technique can be applied to any settings in which documents have overlapping content but different formats, such as corporate press releases and 10-K filings.

In summary, supervised deep learning can scale up human effort in complex, time-consuming tasks. Future efforts in developing or applying advanced topic models may yield more sensible and useful analysis for accounting researchers.

7. Conclusion

We survey publications in top accounting journals and observe a sustained growth of textual analysis as a research method for accounting researchers. We organize available NLP methods in a unified framework and discuss their current and potential applications in accounting. We believe that newer NLP methods, such as deep learning and topic modeling, will likely advance textual analysis in accounting.

Although NLP methods have shown considerable promise in accounting research, it is important that researchers invest in understanding how these tools function. There are several things we can do. To start, we can thoroughly validate text-based measures. Validation not only develops confidence in a measure but also illuminates the measure's characteristics and behaviors. Next, researchers can use available interpretability techniques.[40] Significant computer science research is ongoing to make deep learning models more explainable (Blazek and Lin 2021; Ivanovs et al. 2021). As the techniques expand, accounting researchers can improve their ability to explain what is happening inside the models. Third, because readers are typically unfamiliar with machine learning methods, especially text regressions and deep learning, authors can bridge this gap by providing details of their procedures and explaining the mechanism in plain language.[41] Last, researchers can openly share code, models, and data, especially after publication.

Despite our best efforts to understand how machine learning methods function, they can still be subject to the "it's a black box" criticism. One comfort, however, is that we primarily use NLP tools to construct new proxies (e.g., sentiment and readability), and not to test hypotheses

40. Such as SHapley Additive exPlanations (SHAP) and the integrated gradient method (Sundararajan et al. 2017).
41. For example, text regressions differ drastically from OLS in two aspects: (i) hundreds or thousands of features are used as independent variables and the number of features may exceed the number of observations, and (ii) the text features collectively capture the underlying construct, with some contributing more and others less, whereas in an OLS each independent variable represents a construct.

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directly. Therefore, as long as the measures are validated, they can be used with a fair degree of confidence when combined with the econometric tools we already have to test hypotheses.

As a final point, we remind researchers to avoid the temptation to prioritize the tool over the research question, unless the end goal is a methodological study. Sophisticated and interesting tools can sometimes draw our focus away from important research questions and make us overly enamored with the tool itself. Some research questions cannot be readily answered with textual analysis; other questions may benefit from textual analysis but do not require a sophisticated NLP model. As researchers, we must remember the central importance of the questions we are asking and remember that the tools at our disposal are a means to an end.

Appendix 1: Glossary

Terminology	Definition
activation function	Function used to map inputs to an output of each node in a layer of a neural network. For example, the sigmoid function is a commonly used activation function
annotation	Process of assigning categories, labels, and relationships to unstructured data (e.g., textual data). The terms "annotated" and "labeled" are exchangeable
artificial neural network (ANN)	Collection of layers of nodes connected by edges, where the edges have weights that are usually optimized via machine learning. ANNs are also called neural networks, and the nodes are called artificial neurons. An ANN has one input layer, one or more hidden layers, and one output layer, and uses an activation function to transform the inputs for one layer to its outputs, which are then inputs for the subsequent layer
attention	Model that is aware of long-distance connections among words, coupled with the ability to forget when the context gets far away or becomes irrelevant
backpropagation	Efficient way of minimizing the loss function in training an ANN by starting with the error values of the last layer and then calculating the error values of the layer before the last layer, and so on, from right to left until the first layer. The weights are then updated in such a way that nodes with higher error values receive lower weights
bag of words (BOW)	Text representation approach that represents text as a feature vector where each feature is a binary indicator, word count, or weighted word count
bidirectional encoder representations from transformers (BERT)	Pretrained unsupervised deep learning model that performs bidirectional (both left-to-right and right-to-left) representation. BERT is also a contextualized embedding model
classification	NLP task that assigns categorical values to the output of an observation
classification and regression trees (CART)	Classification trees are used when the output variable is categorical; regression trees are used when the output variable is continuous. The model selects input variables at each level of the tree as well as the split points on those variables until a suitable tree is constructed to capture the relationship between inputs and outputs of the training sample. For a new observation, the model finds the best tree route based on the observation's input characteristics and uses the resulting leaf node value as the predicted output
convolutional neural network (CNN)	Deep learning model where each layer recognizes progressively greater levels of detail and subtlety in the underlying data. CNN was originally designed to process image data but was later applied to natural language
corpus	Collection of documents
deep learning	Artificial neural network that uses multiple hidden layers (typically three or more), where each layer performs a different transformation of its inputs to extract features with multiple levels of abstraction (e.g., low-level in an earlier layer and high-level in a later layer)

(The table is continued on the next page.)

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(continued)

Terminology	Definition
document	Set of words, numbers, and other tokens that collectively form a single observation for use in NLP
ensemble learning	Combining multiple machine learning models trained on the same sample—or bootstrapping the training sample for a single model—to improve the stability and accuracy of predictions. The combination might use the mean, majority rule, or different models in sequence
feature	Individual measurable property or characteristic. For example, the occurrence of a word is a feature. Features are used as inputs to a machine learning model. "Feature" is equivalent to "variable" in traditional statistics
feature engineering	Process of extracting and selecting features from the underlying data
F-score	Performance metric for machine learning. An unweighted *F*-score (i.e., F_1-score) is calculated as the harmonic mean of precision and recall. The *F*-score may be weighted to emphasize either precision or recall
hyperparameters	Model parameters that influence how a machine learning process begins, operates, and converges. Hyperparameters are chosen before the learning process begins and are adjusted by a data analyst during the learning process
K-nearest neighbors (KNN)	Supervised learning algorithm that can be used to produce categorical outputs for classification or continuous outputs for regression. "Nearest neighbors" are the data points in the training set that have minimum distance in the feature space from a new data point. "*K*" is the number of such data points that a data analyst considers in implementing the algorithm
lasso	Least absolute shrinkage and selection operator. It is a popular penalized regression estimation method that can shrink the influence of certain features to zero to address overfitting
lemmatization	Preprocessing procedure in NLP that considers the context and converts a word to its meaningful base form ("lemma"). For example, the lemma of the verb "caring" is "care"
long short-term memory (LSTM)	RNN-based deep learning model. LSTM improves traditional RNNs by addressing the vanishing gradient problem (i.e., the gradient becomes so small that it prevents the weight from changing, thereby restricting further learning)
machine learning	Automated detection of meaningful patterns in data by using algorithms with training data
naïve Bayes (NB)	Probabilistic supervised learning classifier based on applying Bayes' theorem (https://en.wikipedia.org/wiki/Bayes%27_theorem)
named entity recognition (NER)	Information extraction task that locates and classifies named entities (i.e., real-world objects) in text into predefined categories such as locations, organizations, and quantities
natural language processing (NLP)	Subfield of computer science that uses algorithms to process human (natural) language in the form of text or speech
overfitting	Problem when the model learns the detail and noise in the training data so well that it performs poorly on new data
penalized regression	Family of regression estimation methods that addresses overfitting by adding a component to the loss function to shrink the influence of certain features
precision	Rate of positive identifications that are actual positives
pretrained model	Model that has already been trained on a large corpus, thus allowing for less training effort by users

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(continued)

Terminology	Definition
random forest (RF)	Model that draws a bootstrapped sample of the training data and then draws a random sample of the full set of input variables to yield one decision or regression tree. The procedure is repeated to build many trees. The final prediction for an observation in the training sample is the aggregate of all predictions from these trees about this observation
recall	Rate of actual positives that are identified as positives. It is one minus the Type II error rate. Recall is also called "sensitivity"
recurrent neural network (RNN)	Deep learning model with a recurrent connection on each hidden state (i.e., a looping constraint in each node of the hidden layer) to ensure that sequential information in the input data is captured
regularization	Method to penalize models for overfitting
ridge	Penalized regression estimation method that can shrink the influence of certain features to a lower level but not zero
stemming	Rules-based preprocessing procedure used to approximate a word's lemma. For example, after stemming, "expectation" becomes "expect," and "caring" becomes "care." It is more efficient than lemmatization, but less accurate
supervised learning	Machine learning style where the model is trained on inputs and labeled outputs to learn how to map inputs to outputs
support vector machine (SVM)	Supervised learning model that divides the training data into regions separated by a linear boundary, which may be linear or nonlinear with respect to the original feature space. When the output variable is continuous, SVR is used
testing sample	Portion of the sample unseen by the machine during training. It is used to generate performance metrics for evaluation of the final model
token	Individual lexical unit in text data (e.g., word, number, or abbreviation)
training	Process to determine the model parameters that best capture the relationship between inputs and outputs in the training sample
training sample	Sample with input and labeled output for a machine to learn the relationship between inputs and outputs
transfer learning	Approach in machine learning that applies the knowledge gained from one task to a similar task. Transfer learning is the mechanism that allows for pretrained models
transformer	Deep learning model that uses an encoder, a decoder, and an attention mechanism. The encoder reads each input. The attention mechanism, called "soft attention," processes an input sequence, decides which other parts of the sequence are important, and assigns different weights to those inputs. The decoder takes the encoded sequence and weights and generates another sequence
unsupervised learning	Machine learning style where the model is trained only on unlabeled data, letting the model find the structure in the data
vector space model (VSM)	Another name for BOW
word embedding	Text representation approach that allows words with similar meanings to have a similar representation. Recent embedding models use a neural network to map words into vectors of reduced dimensions, where each vector represents a latent semantic meaning. For example, "pleased," "glad," and "happy" have similar vector representations even though the words are distinct and independent under BOW. Word embeddings are also referred to as "word vectors"

Note: These definitions are meant to be functional and accessible to a general audience, not for technical completeness.

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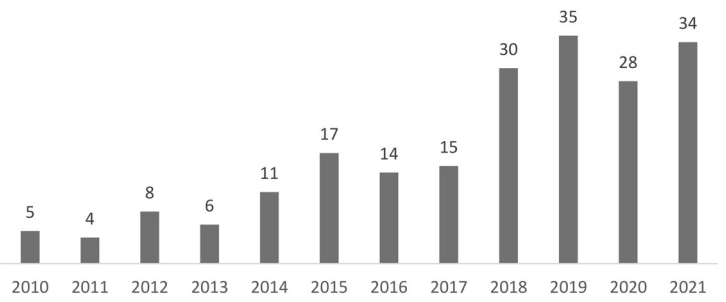
Appendix 2: Survey of textual analysis in published accounting research

To understand the popularity and importance of textual analysis in accounting research, we review studies published in top accounting journals during 2010–2021. These journals are *The Accounting Review* (*TAR*), *Journal of Accounting Research* (*JAR*), *Journal of Accounting and Economics* (*JAE*), *Contemporary Accounting Research* (*CAR*), *Review of Accounting Studies* (*RAST*), *Accounting, Organizations, and Society* (*AOS*), and the accounting track of *Management Science* (*MS*). We downloaded all publications and, for each publication, read the abstract and skimmed the research analysis to identify the initial set of publications that use textual analysis. For the unselected publications, we conducted a keyword search and added them to the initial set if a keyword was found.[42] We then reviewed each selected article to confirm that it uses textual analysis in some fashion.[43] As a result, we collected 207 publications that employ textual analysis. We classified each publication into our categories of interest.[44]

Figure 3 summarizes the publications in seven ways. The number of publications has grown dramatically over time, indicating the increasing importance of textual analysis in accounting research. Even though textual analysis was spurred by financial accounting research in 2008, studies in other accounting research areas have all adopted this research method.[45] Corporate disclosures account for the vast majority of the textual data sources. This observation is unsurprising; corporate disclosures are also the main data source for prominent textual analysis studies in finance and economics (Hoberg and Phillips 2010; Hassan et al. 2019). Textual analysis is most commonly used for creating a key variable for hypothesis testing. Accounting researchers often use textual analysis to capture sentiment (tone), readability, disclosure quantity, disclosure similarity, forward-looking information, and topic discovery.

Figure 3 Textual analysis publications

Panel A: Number of textual analysis publications in top accounting journals by year



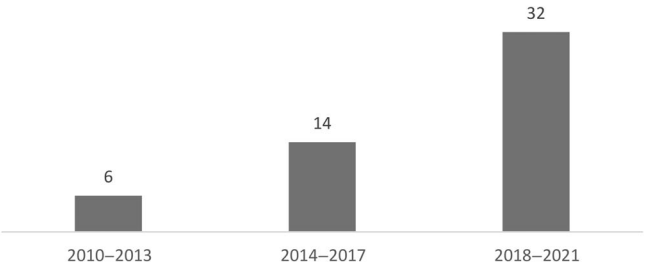
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42. The case-insensitive keywords are "perl," "python," "algorithm," "R," "parse," "text," "tone," "readability," "complexity," "cosine," "search," "extract," "machine," "learn," "bayes," and "train."
43. Automated keyword search without human review would have generated many false positives because some studies discuss textual analysis papers without conducting textual analysis. Our research assistant provided preliminary classifications of the publications. One coauthor systematically classified the publications. Another coauthor reviewed some of the classifications.
44. Except for two studies using experiments and seven studies using field research or other qualitative methods, all publications use the archival research method.
45. F. Li (2008) marked the beginning of the modern era of textual analysis in accounting. In finance, the first study using textual analysis is Antweiler and Frank (2004), who classify messages on Yahoo Finance and Raging Bull into buy, hold, or sell categories, although Tetlock (2007) seems to be the first to coincide with broader popularity.

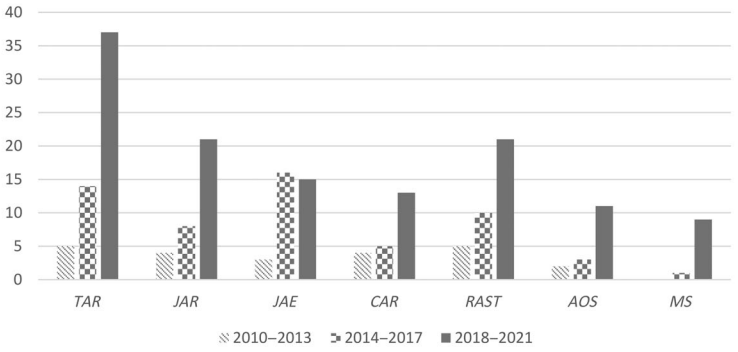
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Figure 3 (continued)

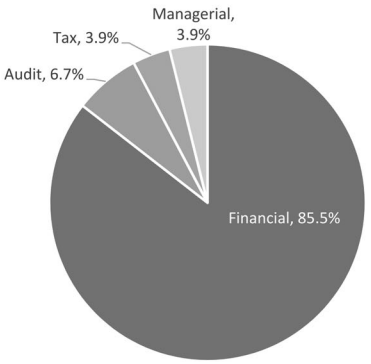
Panel B: Number of textual analysis publications per year in top accounting journals by periods



Panel C: Total number of textual analysis publications in each period by journal



Panel D: Number of textual analysis publications by research area

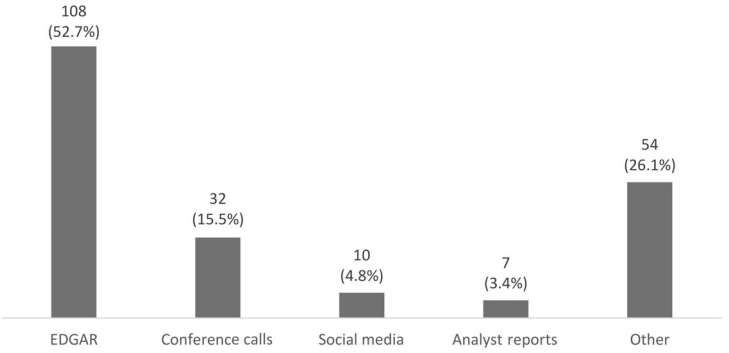


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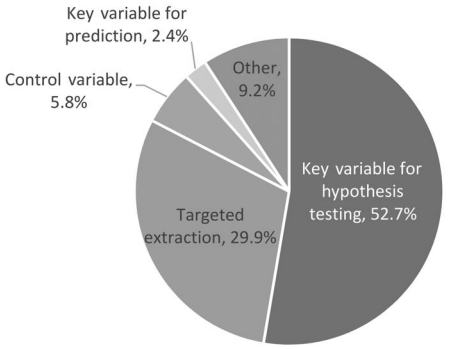
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Figure 3 (continued)

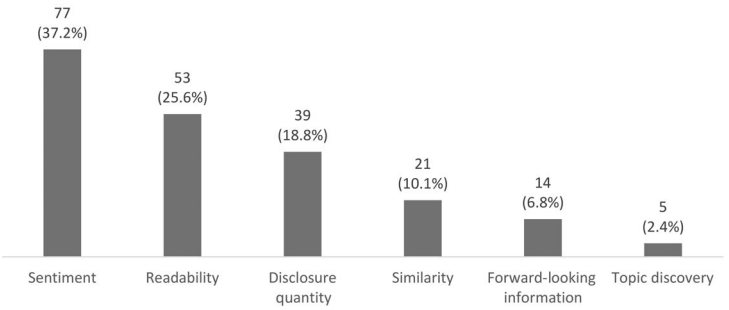
Panel E: Number of textual analysis publications by textual data source



Panel F: The role of textual analysis in a study



Panel G: Number of textual analysis publications that perform common textual analysis tasks



Notes: Panel A reports the number of textual analysis publications in top accounting journals by year. Panel B reports the number of textual analysis publications per year in top accounting journals for the subperiods: the early period (2010–2013), middle period (2014–2017), and recent period (2018–2021). Panel C reports the total number of textual analysis publications for the subperiods by journal. Panel D reports the percentage of the 207 textual analysis publications by research area. A paper is classified as "auditing" if the title of the paper refers to auditors or audit quality. A paper is classified as "tax" if the title includes the word "tax." A paper is classified as "managerial" if the subject of the paper is compensation, corporate governance, management controls, or supplier relationships. Panel E plots the number as well as percentage

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Figure 3 (continued)

of textual analysis publications by data source. Most studies use corporate disclosures filed with the SEC available on EDGAR. The next major data source is earnings conference call transcripts, social media (e.g., Twitter, Glassdoor, Seeking Alpha, and Yahoo message boards), and financial analysts' research reports. Other sources include Lexis/Nexis, Factiva, CSR reports, patent documents, court documents, Kickstarter, interviews, and international sources. A few studies use more than one source and are classified to more than one category. In panel F, we define five roles in decreasing order of importance. The "key variable for hypothesis testing" role means that textual analysis produces a measure directly from sample documents to serve as either the dependent variable or an explanatory variable in hypothesis testing. A "key variable for prediction" role means that textual analysis produces text-based variables as predictors of an outcome variable of interest (e.g., future earnings, fraud, and bankruptcy) absent hypothesis testing. The "targeted extraction" role means that researchers use textual analysis to extract excerpts from textual data and construct a variable (e.g., perceived competition) from the extracted data using either manual coding or computer algorithms. A "control variable" role means that textual analysis directly produces a control variable from sample documents. A role of "other" includes exploratory analysis, supplementary analysis, robustness checks, or methodology. If a study uses textual analysis for more than one role, we categorize the study into the role with the most importance. Panel G summarizes the publications by the type of construct that textual analysis intends to capture, and we refer to the activity of generating an output to reflect that construct as a "task." If a study performs more than one task, it is classified into more than one category. The denominator for the percentages in panels E–G is 207.

Appendix 3: Facets of construct validity

Construct validity can be evaluated under two approaches: translation validity and criterion-related validity (Bloomfield et al. 2016).[46] There are two types of translation validity. Face validity means that, on its face, a measure seems to be a good translation of the construct. For example, if a text-based measure indicates the existence of tax shelters, then that disclosure should contain "tax." Content validity means that the measure captures the conceptual meaning of the construct (Bedford and Spekle 2018). Content validity implies face validity and therefore is the main way to establish translation validity. For example, if textual analysis automates and therefore scales up human assessments of a disclosure's readability, to establish content validity the researcher can compare the text-based measure with human assessments in a small randomly selected sample. This is a direct comparison. The researcher needs to set a threshold ex ante for an acceptable rate of agreement and then decides whether validity can be affirmed. Even though this rate is a judgment decision by researchers, we recommend 80% for binary classifications.

There are four types of criterion-related validity. For convergent validity, researchers check whether the operationalization is closely correlated with other operationalizations that have been claimed to capture similar constructs. For example, a text-based measure for climate risk should have a high correlation with an indicator variable for industries with high exposure to climate risk. For concurrent validity, researchers check whether the operationalization is closely correlated with a behavior that is expected to occur concurrently. For example, a text-based measure for firm-level political risk should have a high correlation with the amount of corporate political spending. For predictive validity, researchers check the operationalization's ability to predict an outcome that will result from the underlying construct. For example, a text-based measure for CEO overconfidence is expected to predict future corporate overinvestment by overconfident CEOs. Donovan et al. (2021) establish the predictive validity of their text-based credit risk measure by showing that their measure is significantly associated with future credit events

46. Also see *Research Methods Knowledge Base*, an e-book written by Professor William M. K. Trochim and hosted at Conjointly (https://conjointly.com/kb/measurement-validity-types/).

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(e.g., bankruptcy). For discriminant validity, researchers check whether the operationalization has a low correlation with other operationalizations that capture quite different constructs. For example, a text-based measure for innovative corporate culture should have a low correlation with the number of corporate board meetings per year, which proxies for board monitoring.

Appendix 4: Implementation guide

Panel A: Data acquisition and preprocessing steps	
Data acquisition	Obtain or download data (often in HTML or txt format) from EDGAR, company websites, media, social media platforms, and so on
Data cleaning	Remove HTML tags, non-text characters, and special symbols (e.g., © and §)
Tokenization	Convert text into text units (e.g., words or sentences) using whitespace and punctuation marks
Document filtering	Delete documents that do not meet the minimum length threshold (e.g., documents incorporated by reference)
Spelling and encoding	Use popular spell checkers to identify and correct spelling mistakes. Convert text characters to proper encoding
Stop words[a]	Remove common or stop words (e.g., "the," "in," and "a")
Stemming or lemmatization[b]	Convert inflected words (e.g., increasing, increases, and increased) to their root form (e.g., increase)
Panel B: Steps for the dictionary approach	
Dictionary selection	Select a dictionary that characterizes the topic of interest (e.g., the dictionary of positive and negative words for sentiment analysis)
Dictionary structure	Learn the dictionary structure (single words vs. phrases, words in the root form, and inflected words) and how individual dictionary entries are coded
Word counts	Determine how individual dictionary entries will be counted (binary count or cumulative frequency count)
Word weights	Determine if all counts are equally weighted or certain words or phrases should receive higher or lower weights (e.g., more common words receiving lower weight). Take into account qualifiers (e.g., "very," "a little," and "much"), negators (e.g., "no," "none," and "neither"), and contrast transitions (e.g., "but," "however," and "although")
Dictionary validation	Compare dictionary performance with human annotators in identifying relevant content. Check the sensitivity of results to excluding individual dictionary terms, adding synonym terms to the dictionary, and using alternative dictionaries if applicable
Scaling	Determine the scalar for the final variable of interest (e.g., the number of total words in the document)
Panel C: Steps for supervised machine learning	
Output variable	Decide how to represent the variable of interest: (i) continuous or (ii) categorical
Sample annotation	Gather an annotated sample (e.g., labeled words, sentences, paragraphs, or articles). Consider annotation platforms (e.g., Prodigy, Amazon Mechanical Turk, TagEditor, SMART, and piaf)
Sample split	Split the annotated sample into subsamples for training, validation, and testing. Ensure that each class of the variable of interest is well represented
Model selection	If a deep learning model is employed, decide on the model (e.g., BERT) and whether to fine-tune the model. If traditional machine learning is used, choose a specific model (e.g., NB, SVM, or RF)
Text representation	Decide whether to represent the text in BOW or word embeddings

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Panel C: Steps for supervised machine learning	
Performance metrics	Decide on the metrics to evaluate model performance. Options include accuracy, precision, recall, *F*-score, and ROC-AUC
Model estimation and evaluation	Fit the model using annotated data, check model performance on the validation data, and determine if more annotated examples are needed. This is an iterative process
Full sample prediction	Use the trained (calibrated) model to predict the outputs of unseen data
Scaling	Determine the scalar for the final variable of interest (e.g., the number of total sentences in the document)

Panel D: Recommended resources	
Purpose	Resources
Basic textual analysis:	nltk: https://www.nltk.org/
Traditional machine learning:	scikit-learn: https://scikit-learn.org/
Python libraries:	Spacy: https://spacy.io/
	Flair: https://github.com/flairNLP/flair
	Stanza/Stanford NLP: https://stanfordnlp.github.io/stanza/
	fastText: https://fasttext.cc/
	TextBlob: https://textblob.readthedocs.io/en/dev/
	Gensim: https://radimrehurek.com/gensim/
	Pattern: https://github.com/clips/pattern
	PyNLPl: https://pynlpl.readthedocs.io/en/latest/
Repositories of NLP models:	Hugging Face: https://huggingface.co/transformers/installation.html
Additional resources:	
Dictionaries	Loughran-McDonald repository: https://sraf.nd.edu/
Data library	Hoberg-Phillips: https://hobergphillips.tuck.dartmouth.edu/
Python tutorial	Anand et al. (2020) and www.textart.us
FinBERT libraries	A. Huang et al. (2022): https://github.com/yya518/FinBERT
Other data sets or code	Conference call measures: https://joshualeeacct.wixsite.com/joshualee/data
	Conference call linguistic complexity: https://danieltayloranalytics.com/data/
	MD&A year-over-year similarity: https://www.stephenvbrown.com/
	Downloading tweets: https://www.allenhuang.org/coding.html
	Corporate culture: https://github.com/MS20190155/Measuring-Corporate-Culture-Using-Machine-Learning

Notes: [a]This procedure is not needed for the NLP methods that use tf–idf weighting or deep learning. [b]This procedure is not needed for the NLP methods that use deep learning or word embedding text representation.

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